

MUSCLE STRUCTURE AND FUNCTION

Introduction.

Muscles are the contractile tissue which provide mobility and stability.

Mobility: Muscles serve a mobility function by producing or controlling the movement of bony lever around a joint axis.

stability: provide stability function by constraining movement of joint surface through approximation of joint segments.

Elements of muscle structure

It is composed of muscle tissue (contractile) and connective tissue (noncontractile).

- * Muscle tissue has the ability to develop tension in response to chemical, electrical or mechanical stimuli.
- * Connective tissue on the other hand, develop tension in response to passive loading.

Composition of a muscle fiber.

* Contractile proteins:

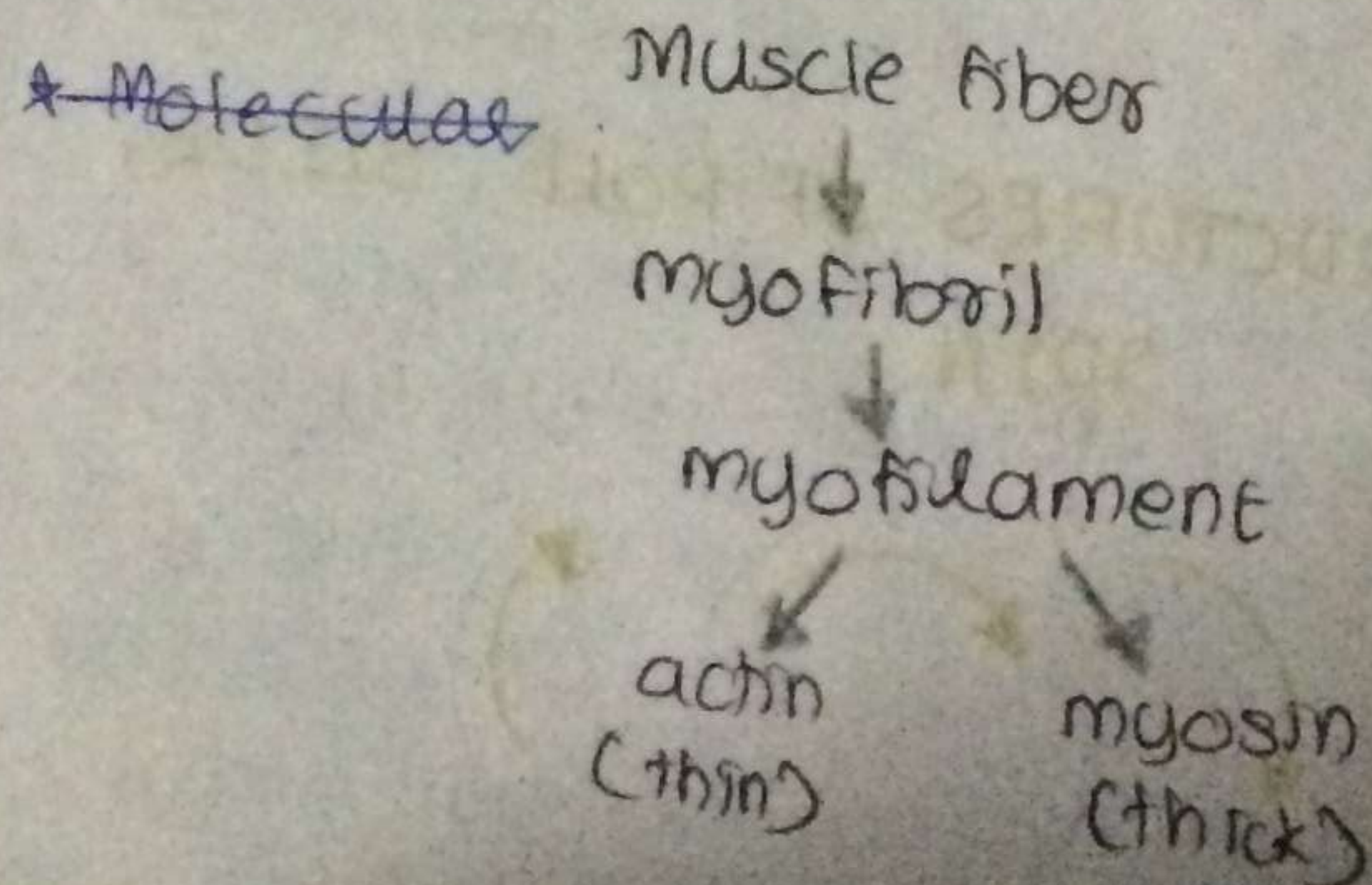
- * Thousands of muscle fibers are grouped into fascicles.
- * Each fascicle is composed of numerous myofibrils.
- * Each myofibril is composed of numerous stacked myofilaments.

Muscle Fiber

↓
fascicles

↓
Muscle

- * The arrangement, number, size and type of the muscle fibers may vary from muscle to muscle.
- * Every single muscle cell is enclosed in a cell membrane called sarcolemma.
- * Cytoplasm of muscle fiber is known as sarcoplasm.
- * Sarcoplasm of muscle fiber contains contractile structures (myofibrils) as well as non myofibrillar structures such as mitochondria, ribosome, glycogen required for cell metabolism.
- * A muscle fiber contains many myofibrils within the fiber.
- * Each myofibril is composed of thick (myosin) and thin (actin) myofilaments formed by actin, myosin, troponin and tropomyosin.
- * The interaction of actin and myosin molecule is essential for muscle contraction.



* Mole
tropo
actin
is att
molec
* The
cule
and

* Stru
* also
* Inclu

* Desmi
force

* Titin
→ It

→ Me

dur

→ ac

Bone

↑

↓

te

con

Actin

Myosin

Actin

* Molecules of the globular protein tropomyosin are found in between two actin strings, and protein tropomyosin is attached to each tropomyosin molecule.

* The tropomyosin and tropomyosin molecule control the binding of actin and myosin molecule myofibril.

* Structural proteins

* also known as non-contractile protein

* Includes Desmin and Titin.

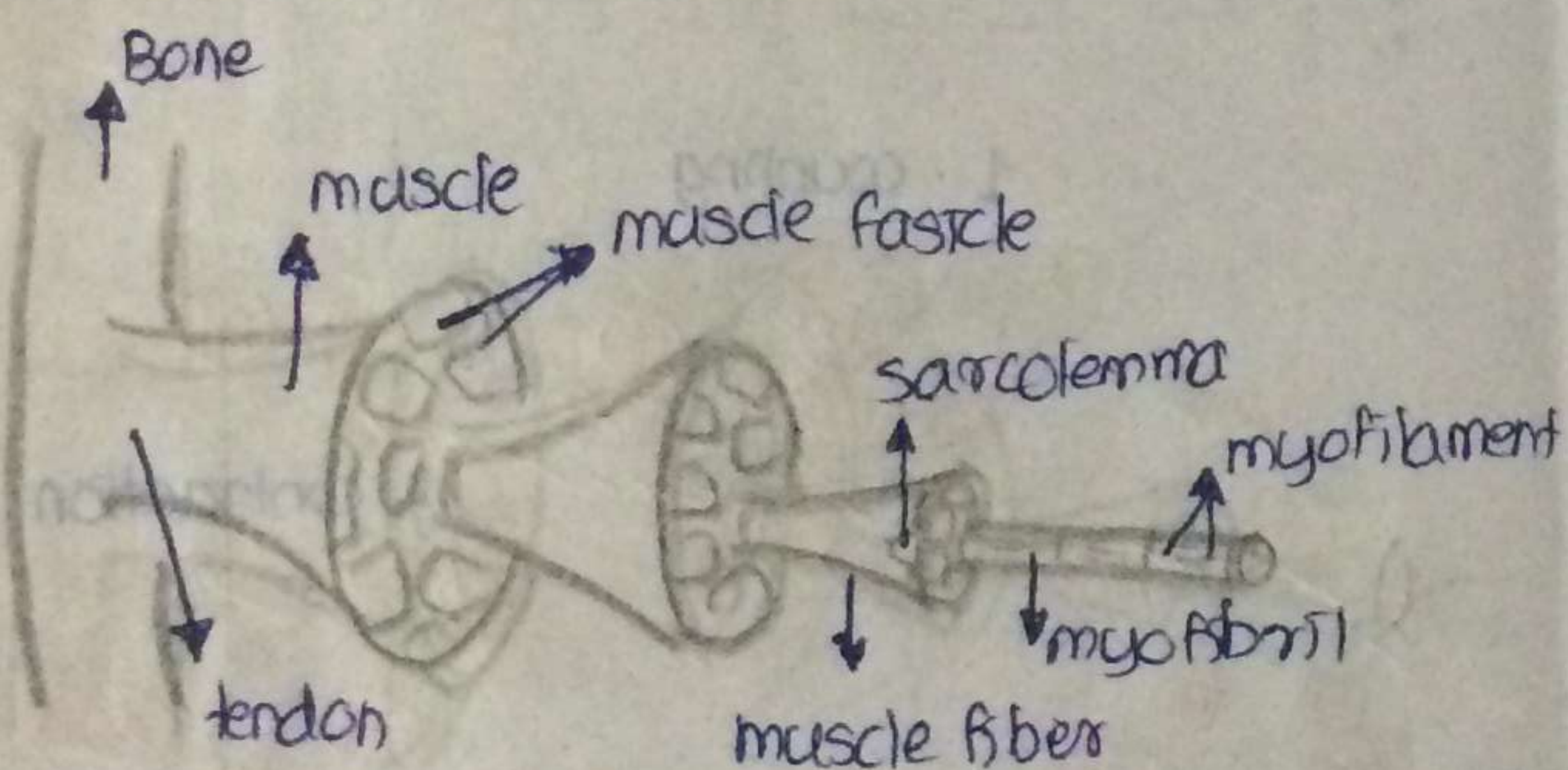
* Desmin helps in transmission of the force along the fiber.

* Titin and desmin eg:

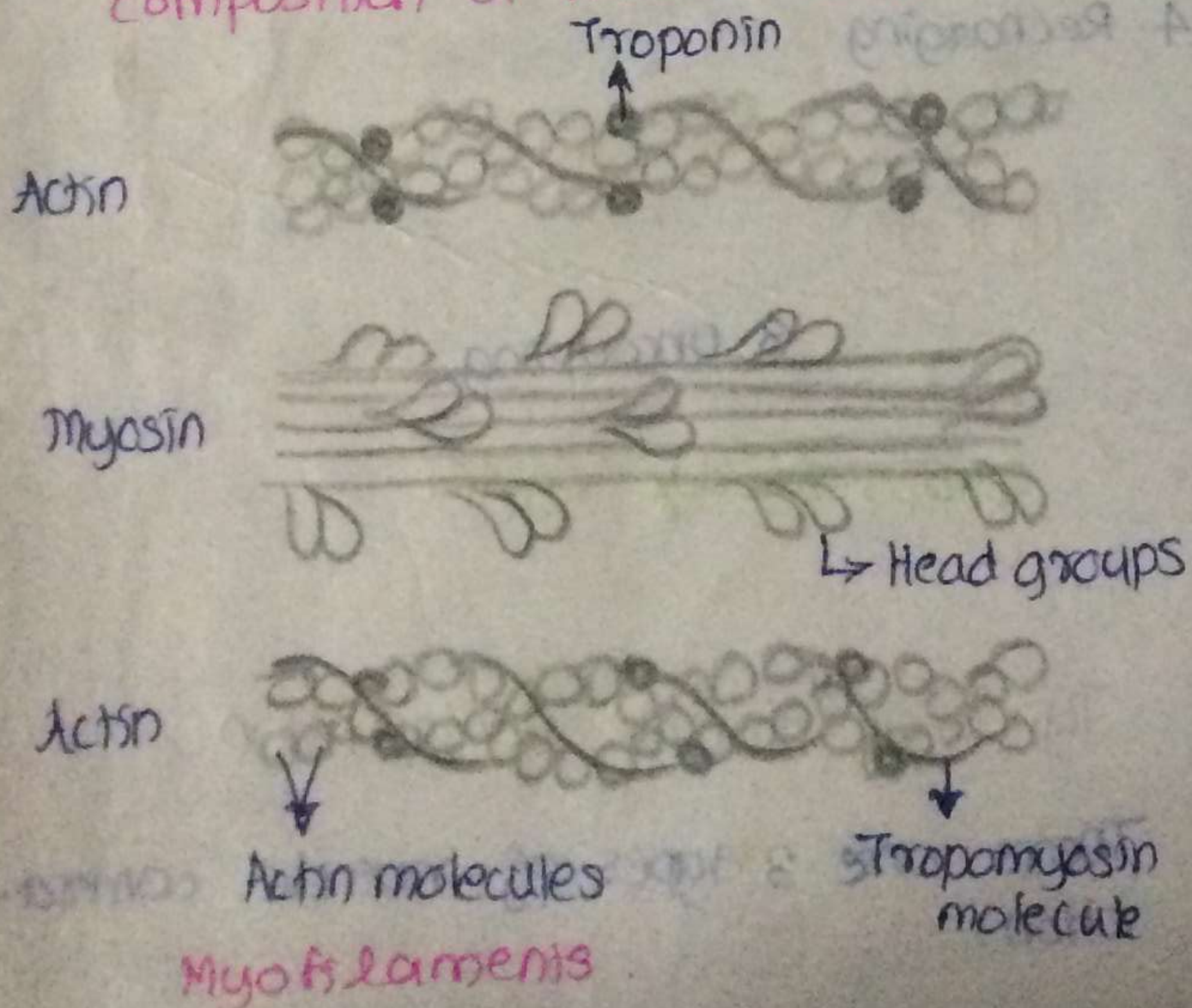
→ It provides structural integrity.

→ Maintain position of thick filaments during muscle contraction.

→ acts as intermediate filaments.



composition of muscle fiber



Contractile unit

* The sarcomere is the contractile unit of the muscle and is distinguished by bands and zones that help to identify the arrangement of thick and thin filaments.

* The boundary of sarcomere is distinguished by Z lines.

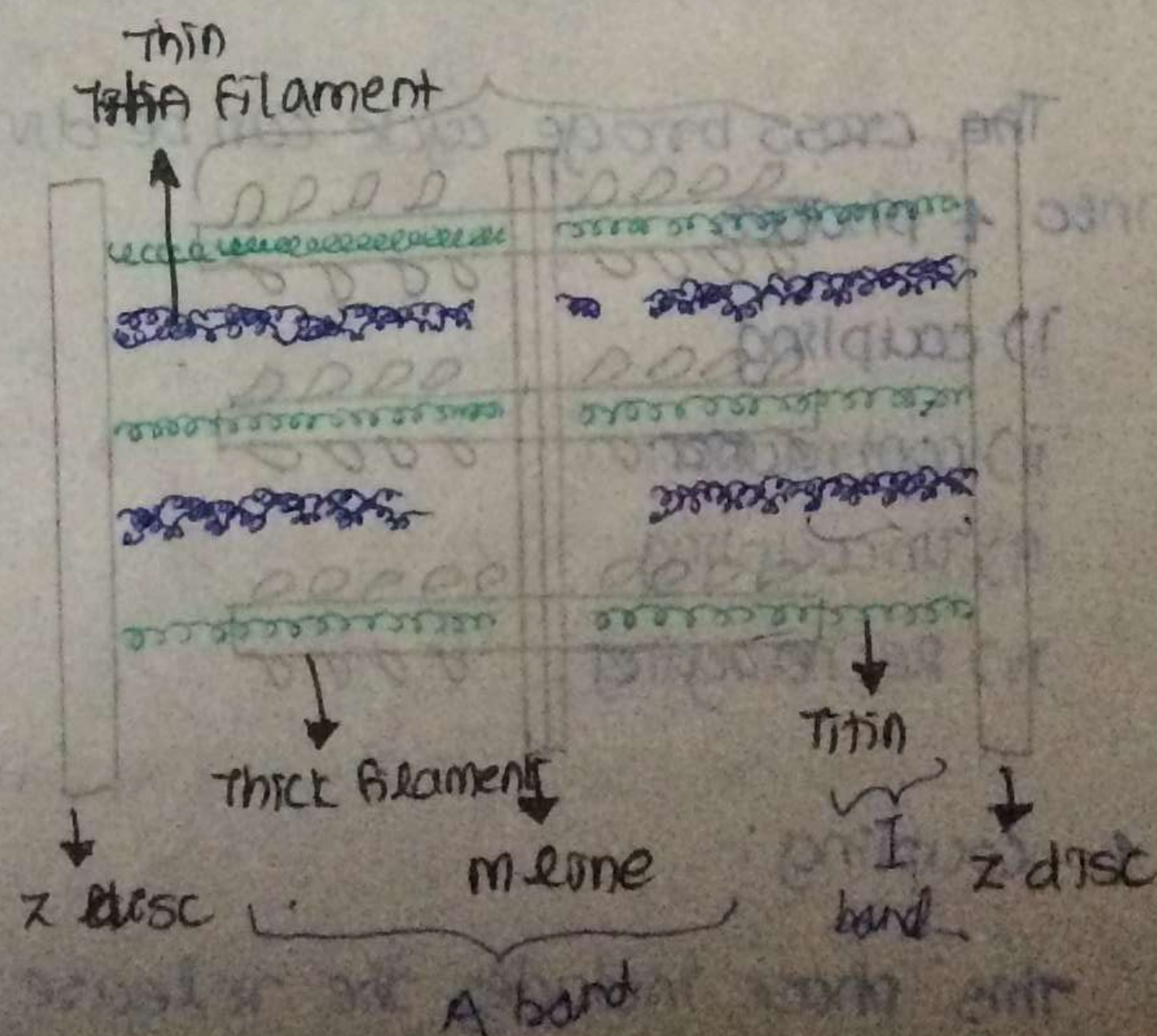
* portion of myofibril located between two Z line is called sarcomere.

* The portion of sarcomere that extends over both the length of thick filaments and small portion of thin filament is called anisotropic or A band.

* Areas that include only actin filament are called isotropic or I band.

* Myosin filament has a central area where there is a no overlap with the thin filaments, called the H zone.

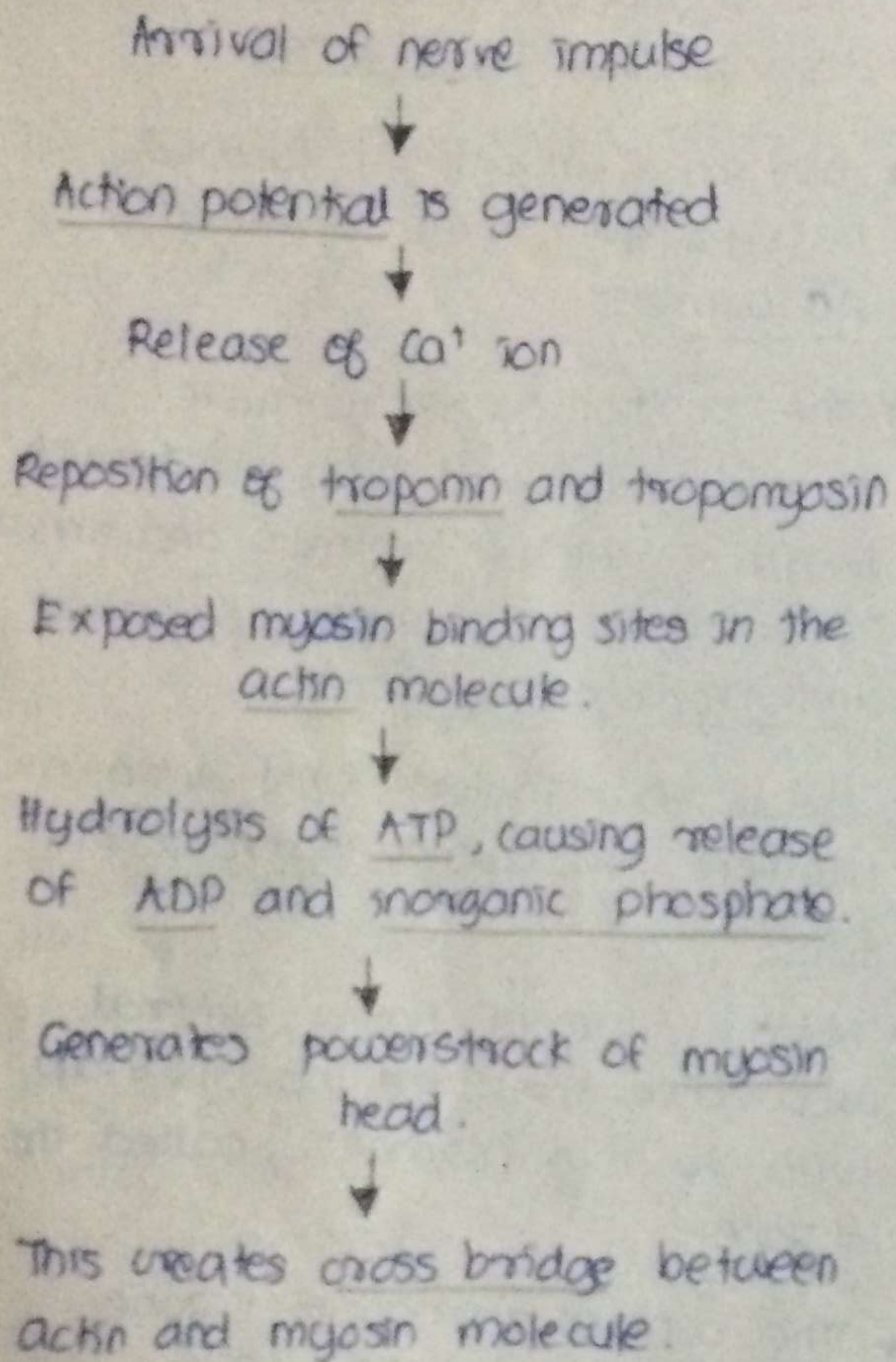
* The central portion of H zone, which consists of the wide middle portion of thick filament is called the M line.



Cross Bridge Interaction

* A muscle contraction is initiated by the interaction of thick and thin filaments within each sarcomere.

* Steps:



Phases of cross bridge cycle

The cross bridge cycle can be divided into 4 phases.

- i) coupling
- ii) contraction
- iii) uncoupling
- iv) recharging

1. coupling

this phase includes the release of

the Ca^{2+} ion and reposition of tropomyosin and tropomyosin.

2. contraction

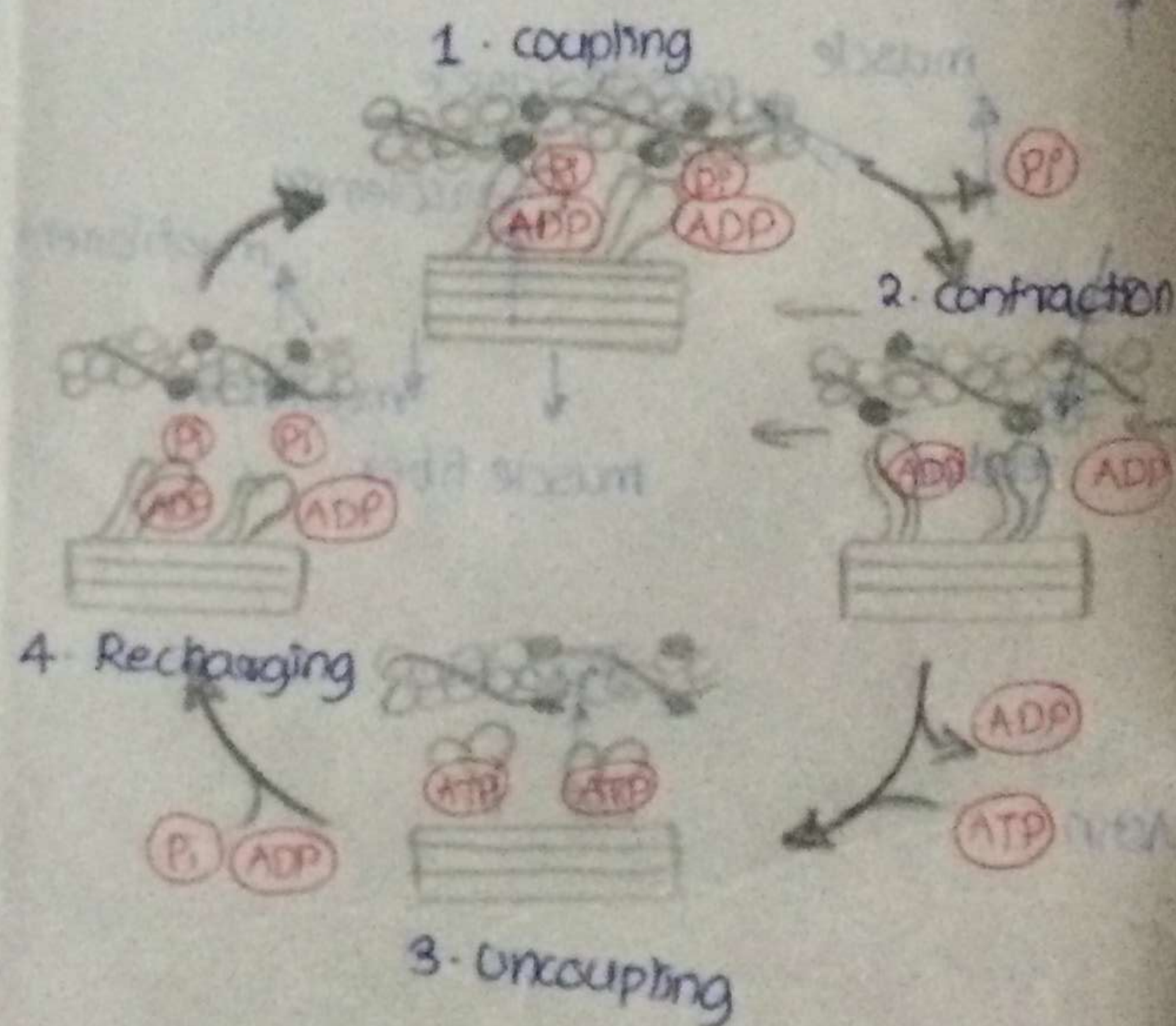
Hydrolysis of ATP (adenosine triphosphate) to release ADP (adenosine diphosphate) + inorganic phosphate, which leads to creation of power stroke in the myosin head.

3. Uncoupling

Binding of new ATP molecule and release of the formed cross bridge. By release of thick filament from thin filament.

4. Recharging

Hydrolysis of ATP and the same cycle is repeated.



Cross Bridge cycle.

Types of muscle contraction

There are 3 types of muscle contraction.

- i) concentric contraction.

ii) Eccentric contraction

iii) Isometric contraction

1. Concentric contraction

- * Also known as shortening contraction.
- * Sarcomere length decreases.
- * Distance between Z line decreases.
- * Length of the muscle also gets decreased.

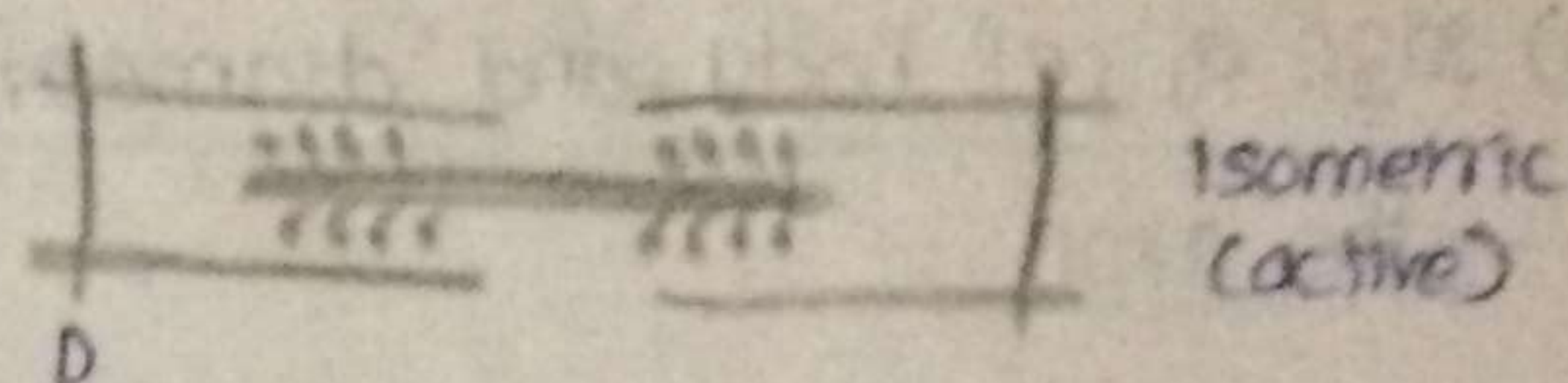
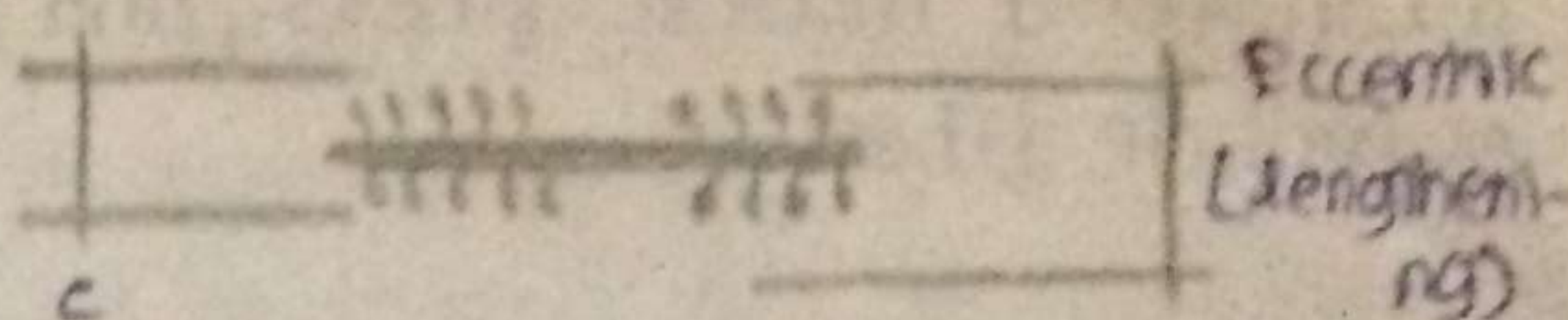
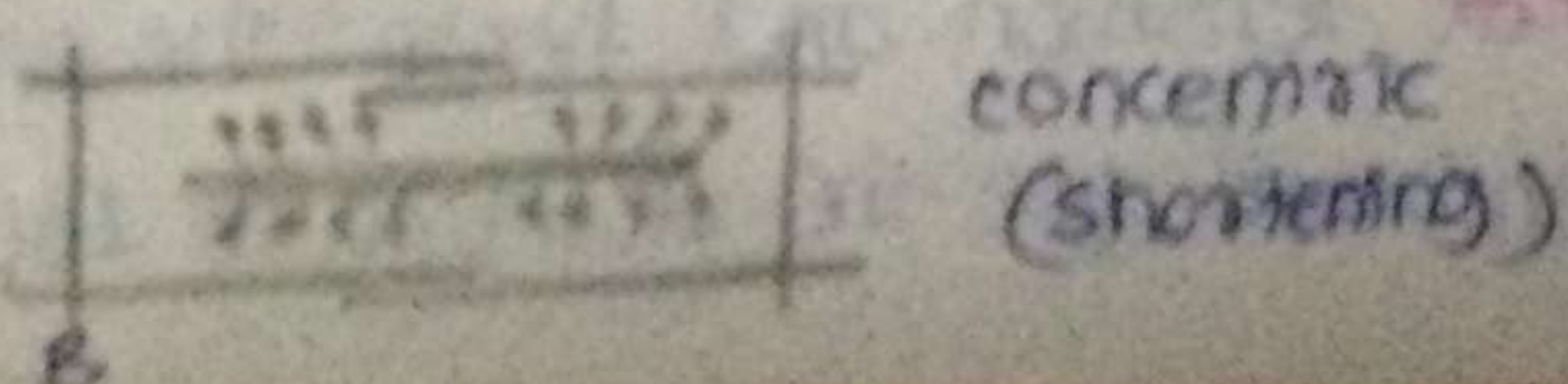
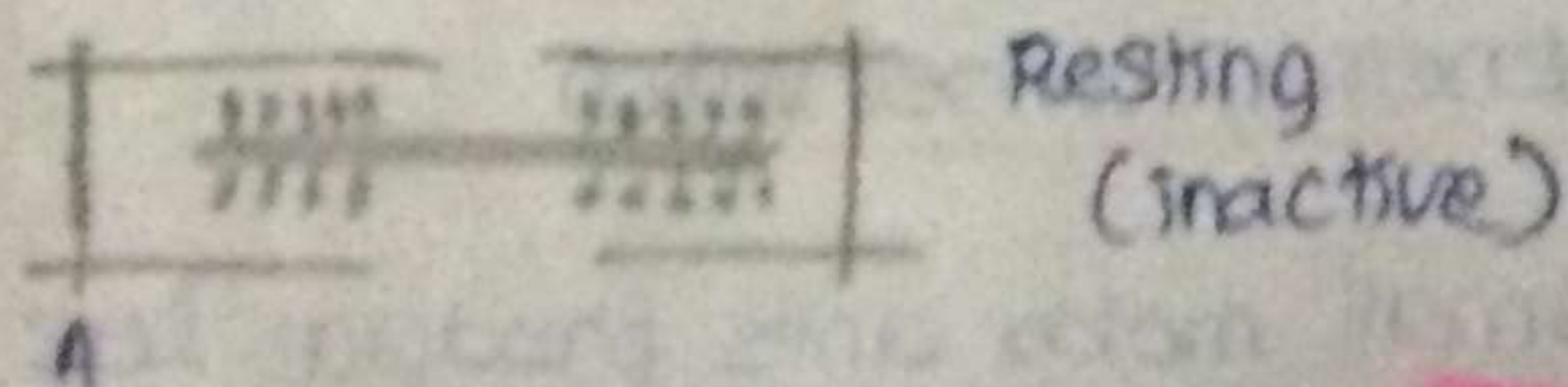
2. Eccentric contraction

- * Also known as lengthening contraction.
- * Sarcomere length is greater than the resting length.
- * Thin filaments are pulled away from thick filaments.
- * Cross bridges are broken.
- * Muscle length increases.
- * Mostly due to external forces.

3. Isometric contraction

Here the sum of forces is zero. i.e., the force generated inside the muscle is equal to externally applied force.

- * contraction occurs without change in length.



MOTOR UNIT

- * The motor unit consists of α -motor neuron along with its cell body and axon and the muscle fibers it innervates.
- * The contractile ~~protein~~ process is initiated when the muscle fibers of a motor unit receive a stimulus transmitted through an α -motor neuron.
- * The cell body of the neuron is located in the ventral horn of spinal cord.
- * The nerve cell axon extends from the cell body to the muscle, where it divides into several small branches.
- * Each small branch terminates in a motor endplate, which lies in close approximation to the sarcolemma of a single muscle fiber.
- * The contraction of the muscle may be a result of synchronised or unsynchronised action of the motor unit.
- * The no. of motor units in a muscle, as well as the structure of these units, varies from muscle to muscle.

VARIATIONS IN MOTOR UNIT

- * Motor units vary according to the size of the neuron cell body, diameter

axon, no: of muscle fibers, and type of muscle fibers.

1) Size of cell body and diameter of the axon.

- * Some motor units may have small cell bodies and others may have large cell body.
- * Small cell bodies have small diameter axon is small, the conduction is slow.
- * If the axon is of large diameter, the conduction and muscle action would be faster.
- * In small diameter unit axon a nerve impulse would take long time to reach the muscle fiber.

2) No: of muscle fibers.

- * Size of motor unit is determined by no: of muscle fibers that it determines.
- * No: of muscle fibers may vary from two to a few thousand.
- * If the muscle is controlling the fine movements, muscle will have small no: of motor unit as well as less no: of muscle fibers per motor unit.
- * If the muscle is controlling the complex movements, the no: of muscle fibers per motor unit would be larger.

1 eg: Muscle that control eye movements contain usually 9 muscle fiber per motor unit.

2 eg: Gastrocnemius muscle contain about 2000 muscle fiber per unit.

NOTE:

Some muscles have less no: of fibers per motor unit. But in them the total no: of motor units may be high. (Continuation of superficial fascia)

eg: platysma: 25 fibers per motor unit, the total motor unit is 1000.

eg: Gastrocnemius: no: of fibers per unit is 2000. But total no: of the motor unit is only around 600.

RECRUITMENT OF MOTOR UNITS

It states that motor units are recruited in order of size from the smallest to the largest depending upon intensity.

SIZE PRINCIPLE: "It states that as more force is needed motor units are recruited in a precised order according to the magnitude of their force output."

- * Usually small units are recruited first followed by large unit.
- * The recruitment of motor unit depends upon 3 things.
 1. Energy conservation
 2. Complexity of the task
 3. Type of muscle action.

1. Energy conservation

Small motor units produce less force or tension and hence the energy expenditure for the activity is low.

- * whereas large scale movements requires high energy expenditure.
- * The body recruits motor unit based on this strategy of energy conservation.

2. Complexity of task.

- * If the task is demanding or high more no: of motor units will be recruited.
- * Fine movements will require less motor unit recruitment.

3. Muscle action.

- * The recruitment of motor units may vary according to the type of the muscle action.
- * ~~Isometric~~ Isometric contraction would require only less no: of motor units as compared to concentric and eccentric.

NOTE:

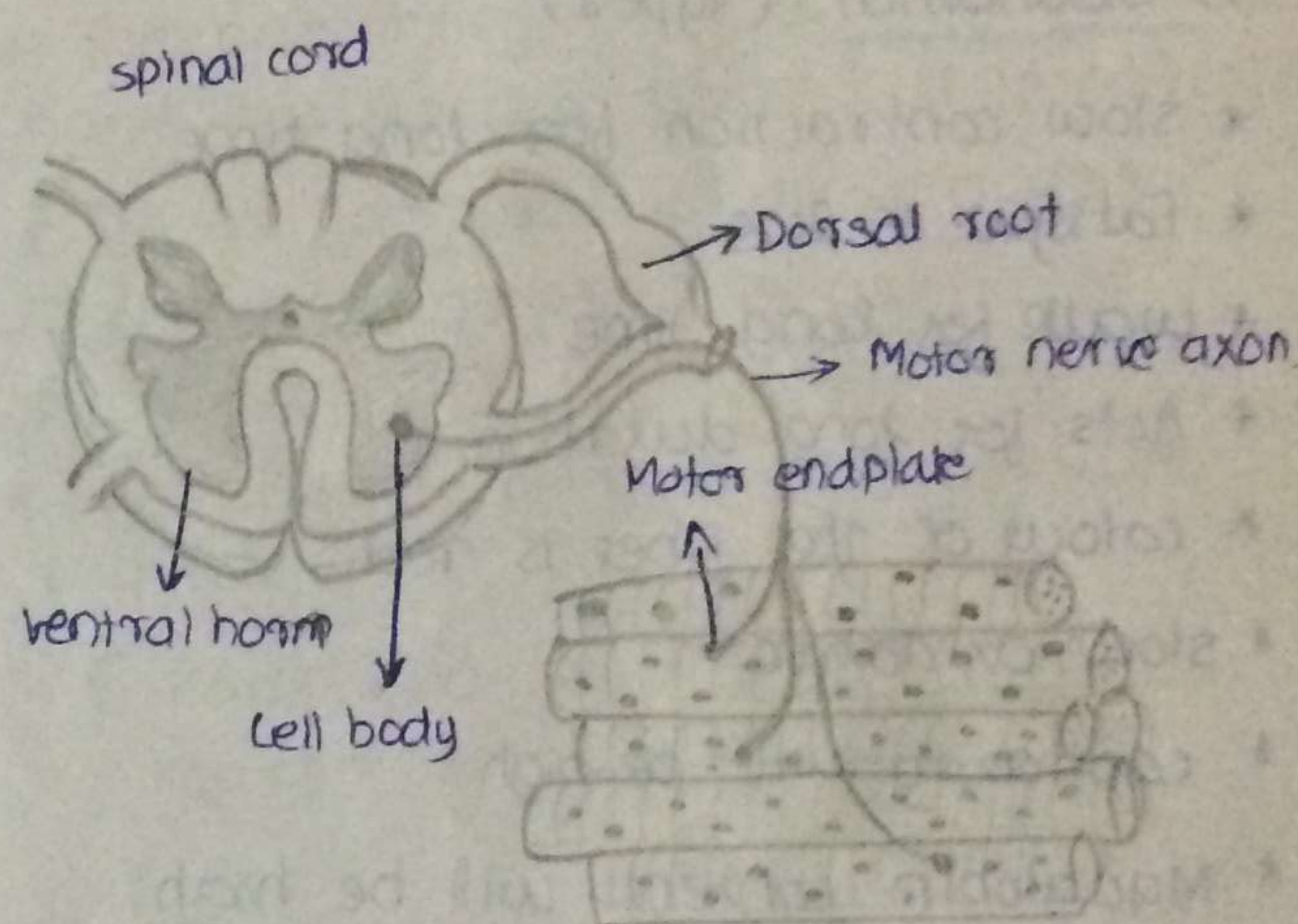
- * The frequency of motor unit firing also effects the force generation.
- * If more no: of motor units are fired, then the force generated will be high.
- * The firing frequency may be different depending on the muscle and its action.

1eg: small distal muscles - depends on ↑ speed frequency of firing. (speed of firing ↑ sed).

2eg: large proximal muscle - decreased frequency of firing (speed of firing is ↓ as no: of motor unit is ↑ sed as there is more no: of motor units to help).

Applications

1. Activities that require longer duration of muscle contraction, the strategy of the body would be to recruit more no: of motor units.
2. In high velocity movements - the firing rate would be higher. Thus there is more chances of injury.



MUSCLE STRUCTURE

1. Fiber type

Muscle fibers in the human body can be of 3 main types:-

- Type I: slow twitch / slow oxidative (SO)
- Type II A: intermediate twitch / fast oxidative glycolytic (FOG)

ii) Type IIx : fast twitch (fast glycolytic anaerobic)

Factors on which composition of fiber type depends.

- i) Genetic factor and race
- ii) Age
- iii) Type of muscle action / muscle function.
(postural, movement producing)

Peculiarities

i) slow twitch (Type I)

- * slow contraction for long time.
- * Fatigue is less
- * work for long time
- * Acts for long duration
- * colour of the fiber is red
- * slow oxidative.
- * capilarity will be high.
- * myoglobin content will be high
- * speed of contraction is low.
- * Resistance to fatigue is high.
- * oxygen rich.

ii) Intermediate twitch (Type II A)

- * Intermediate twitch (Type II A)
- * other name - fast oxidative glycolytic
- * Muscle contraction is of intermediate pace.
- * Myoglobin content is intermediate
- * capilarity intermediate.
- * Mitochondria intermediate
- * Red in colour
- * More prone to fatigue and slow oxidative.

iii) fast twitch : (Type IIx)

- * Type IIx
- * fast glycolytic oxidative
- * Capilarity less
- * Myoglobin amount is less
- * Mitochondria is less
- * White in colour
- * fatigue is high
- * force production is high.
- * O₂ richness is low
- * Usually recruited after type I

	Type I	Type II A	Type IIx
* Diameter	* small	* Intermediate	* large
* Muscle colour	* Red	* Red	* white
* Capilarity	* Dense	* Dense	* Sparse
* Myoglobin content	* High	* Intermediate	* low
* Speed of contraction	* slow	* fast	* fast
* Rate of fatigue	* slow	* Intermediate	* Fast.

2. Muscle Architecture

Based on the muscle architecture, muscle can be classified based on the size, arrangement and length.

Determination of muscle function cannot be based on a single characteristic, it is based on the combination of following characters.

- i) Length of muscle fiber
- ii) ~~len~~ Muscle length
- iii) Muscle mass
- iv) PCSA
- v) amount of pennation.

1. Muscle fiber and muscle length

- * It depends on no: of sarcomere.
- * The more the no: of sarcomere, the fiber can shorten more and produce more amount of force.
- * Thus a long muscle fiber with more sarcomere is capable of shortening more than a short muscle fiber i.e., the force produced is higher.

eg: If a muscle fiber is capable of shortening to 50%, a 6cm long fiber can reduce to 3cm, whereas 4cm long muscle can only reduce to 2cm.

Hence, the longer muscle fiber moves the long lever to greater distance.

2. Muscle mass

Increase in muscle mass or muscle bulk increases the force generating

capacity of muscle.

3. PCSA

Physiological Cross Sectional Area

- * It is the measure of cross-sectional area of muscle perpendicular to the orientation of muscle fiber.
- * The higher the PCSA, the force producing capacity of muscle would be higher, due to increased muscle bulk, increased fiber length and due to more no: of sarcomere.

eg: quadriceps muscle has larger PCSA and hence higher force generating capacity.

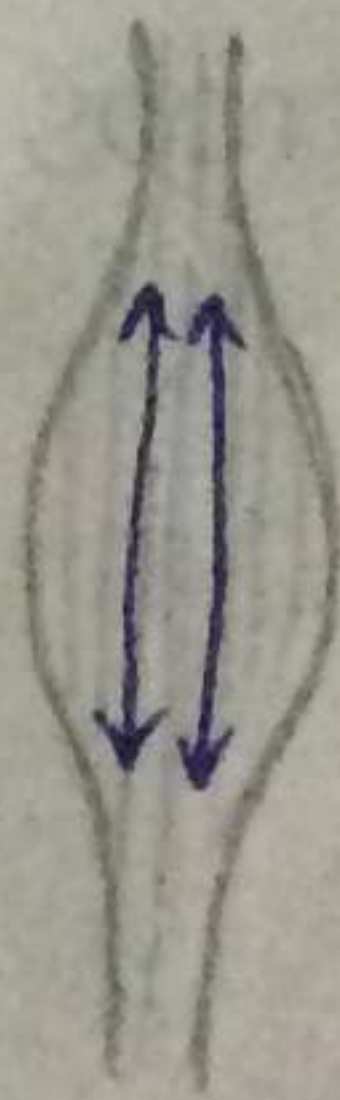
- * The hamstring muscle on the other hand has longer fibers but smaller PCSA as compared to quadriceps. Thus quadriceps acts as ROM provider.
- * Whereas hamstring acts as ROM provider.

(Hamstring: eg of 2 joint muscle, hip extension, knee flexion)

4. Amount of pennation

Based on the degree of pennation muscle can be classified into 4.

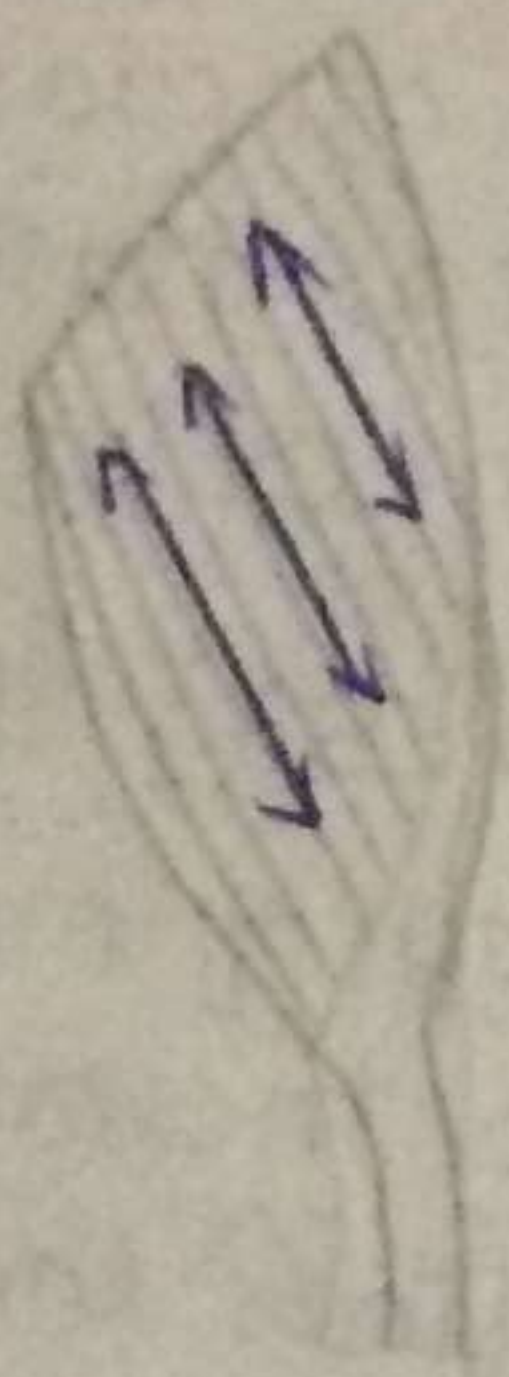
1) fusiform / strap



- * Muscle fibers are parallel to each other.

eg: (SCL) sterno cleido mastoid,
sartorius.

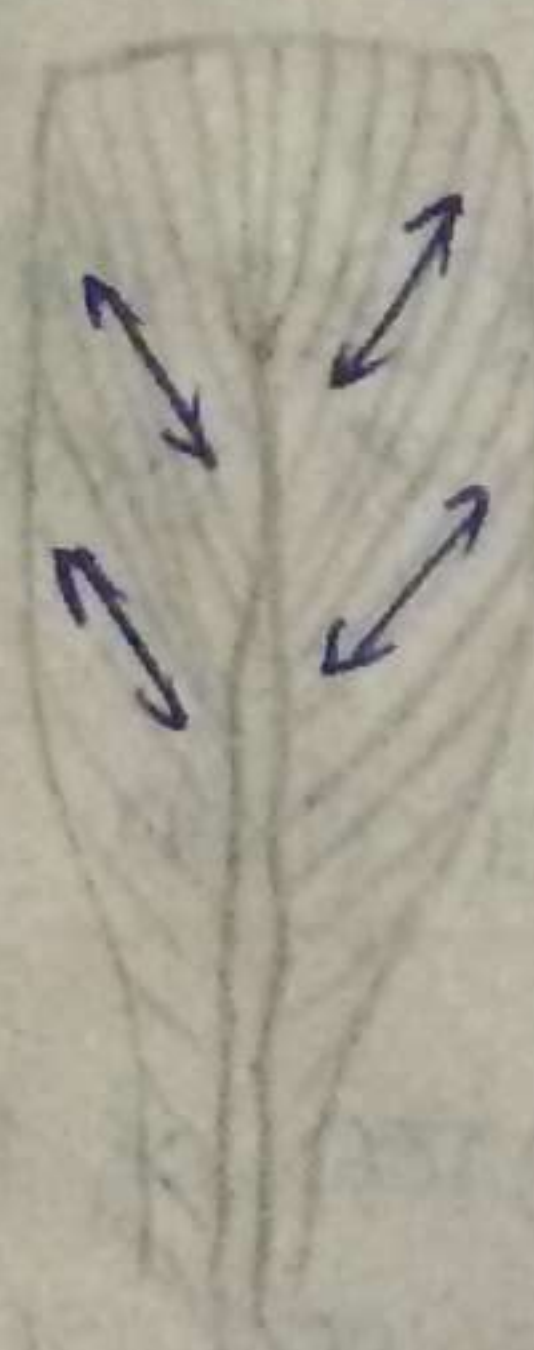
ii) Unipennate



- * Muscle fibers are oblique to the long axis of muscle
- * Fiber length in the pennate muscle are smaller as compared to fusiform muscle.

eg: palmar interossei, lambdoid.

iii) Bipennate



- * Here muscle fibers are arranged obliquely on both sides of a central tendon.

eg: flexor pollicis longus,
Rectus femoris

iii) Multipennate



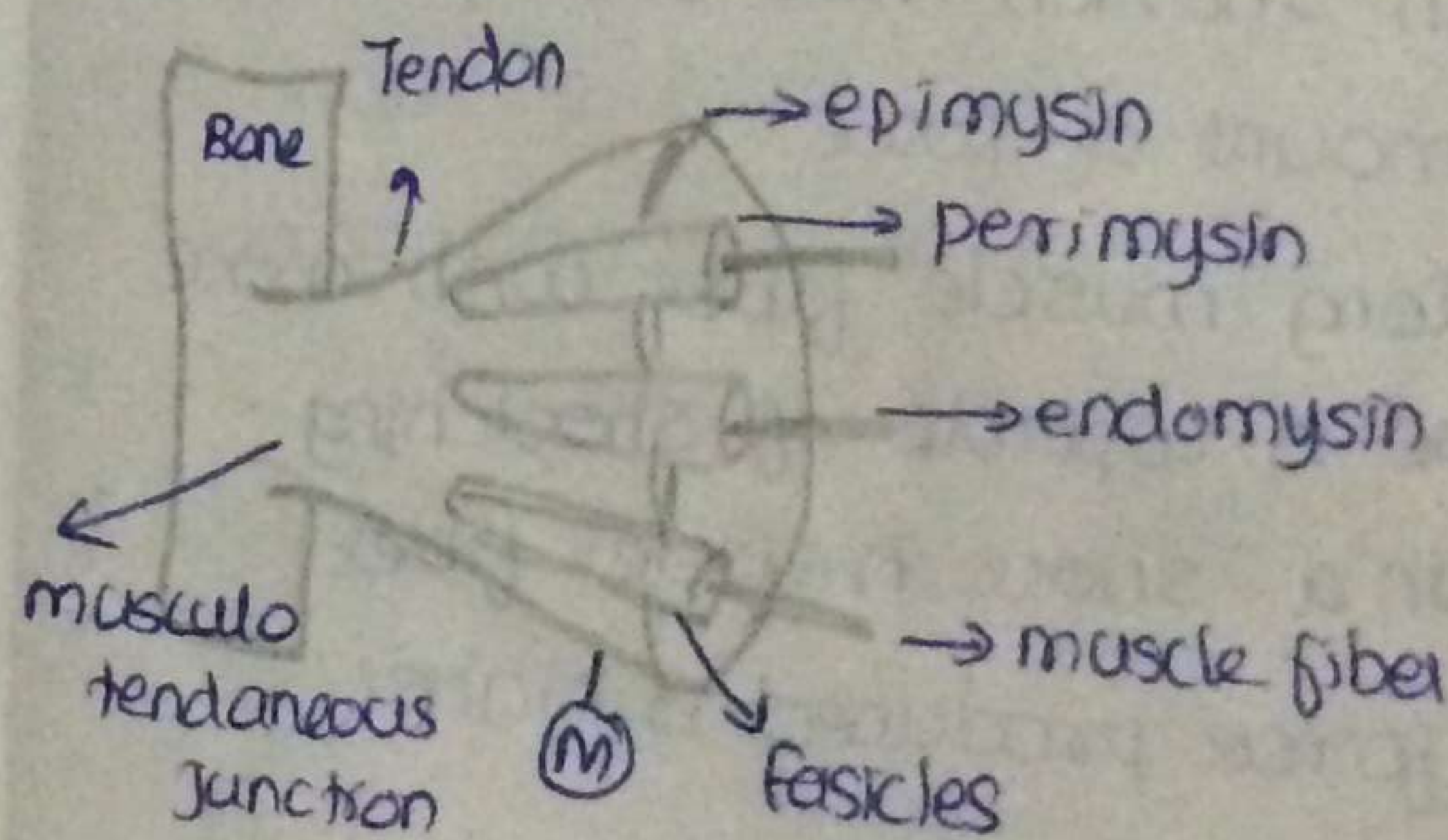
* Here, muscle fibers are obliquely attached to central tendon

eg: Deltoid, soleus muscle.

NOTE

- * Due to pennation the force producing capacity of muscle decreases
- * It is because due to the oblique arrangement only a portion of force is directed to the bony lever but usually the force producing capacity of pennate muscles are not much less as compared to fusiform muscles.
- * It is because as a compensation to loss of muscle force due to pennation in this muscles, no. of muscle fibers are usually higher.

MUSCULAR CONNECTIVE TISSUE



1. Epimysium

The connective tissue that covers the whole muscle. It is formed by continuation of endomysium and perimysium.

2. Perimysium

The connective tissue sheath covers each muscle fascicle.

3. Endomysium

sheath covers each (M) fiber.

MUSCULOTENDANEOUS JUNCTION

- * It is a connection between connective tissues of tendon and muscle fibers.
- * Tendon attaches to muscle at musculotendaneous junction.
- * The attachment is through highly folded membrane where muscle fibers interdigitate with tendon.
- * Here, endotendon, epitendon blend with the epimysium and endomysium of the muscle.
- * The MTJ is seen as a meshwork of interdigitating connective tissue.
- * The MTJ is highly sensitive to mechanical loading and becomes less folded and weaker as load increases.
- * Hence in cases of immobilization, changes occur in the MTJ at the earliest.
- * Hence it is subjected to injury.
- * Hence in rehabilitation the loading should be progressive.
- * Even though MTJ is highly stable, it is the most common site of injuries.

Other connective tissues associated with the muscle.

1. Fascia
2. Aponeurosis
3. Tract / band / sheath
4. Retinaculum.

1. Fascia

- * It is a connective tissue that covers the entire body.
- * It is divided into 2 zones.

i) Superficial:

- * Directly attached to skin
- * Functions are: mobility, insulator, contain muscle

eg: platysma

ii) Deep fascia:

- * Attaches to muscle and bone
- * It is the deep fascia that transform itself into tract / band / sheath or retinaculum.

eg: IT band (Iliotibial band)

It is

2. Tract

- * It is the continuation of deep femoral fascia.
- * The bands acts to transmit the pull of muscle to the bone.

3. Retinaculum

- * Transverse thickening of deep fascia.

eg: Flexor retinaculum of the hand.

4. Aponeurosis

- * These are sheet of fibrous tissue that take place of tendon in flat muscles having wide area of attachment.

eg: Biceptal aponeurosis.

LENGTH - TENSION RELATIONSHIPS OF MUSCLE

→ Tension is of 2 types :- i) passive
ii) Active

Parallel and series elastic components of the muscle.

* All connective tissues inside the muscle contribute parallel and series elastic components.

* 1. parallel

→ the connective tissues that surrounds the muscle i.e., sarcolemma, epimysium, endomysium, perimysium, elastic protein titin and other structures like nerves and blood vessels together forms parallel elastic components of the muscle.

* When a muscle lengthens or shortens the tissue also lengthen & shorten because they function in parallel with the muscle contractile unit.

eg: 1

When a muscle undergoes concentric contraction, the parallel components (collagen) gets shortened.

i.e., During concentric contraction the sarcomere unit gets shortened and together with it the collagen fibres also gets shortened.

The net tension inside the collagen is zero.

eg: 2

During eccentric contraction, the sarcomere gets lengthened and together with it the collagen also gets lengthened, the tension inside the collagen increases.

The increased tension acts as a checking force to prevent further

motions.

→ The rise or fall in passive tension affects the total tension output of the muscle.

$$(\text{total tension} = \text{active} + \text{passive})$$

2. series

* Tendon is the primary structure that acts as series elastic component.

* When a muscle is at rest, the tendon is relaxed.

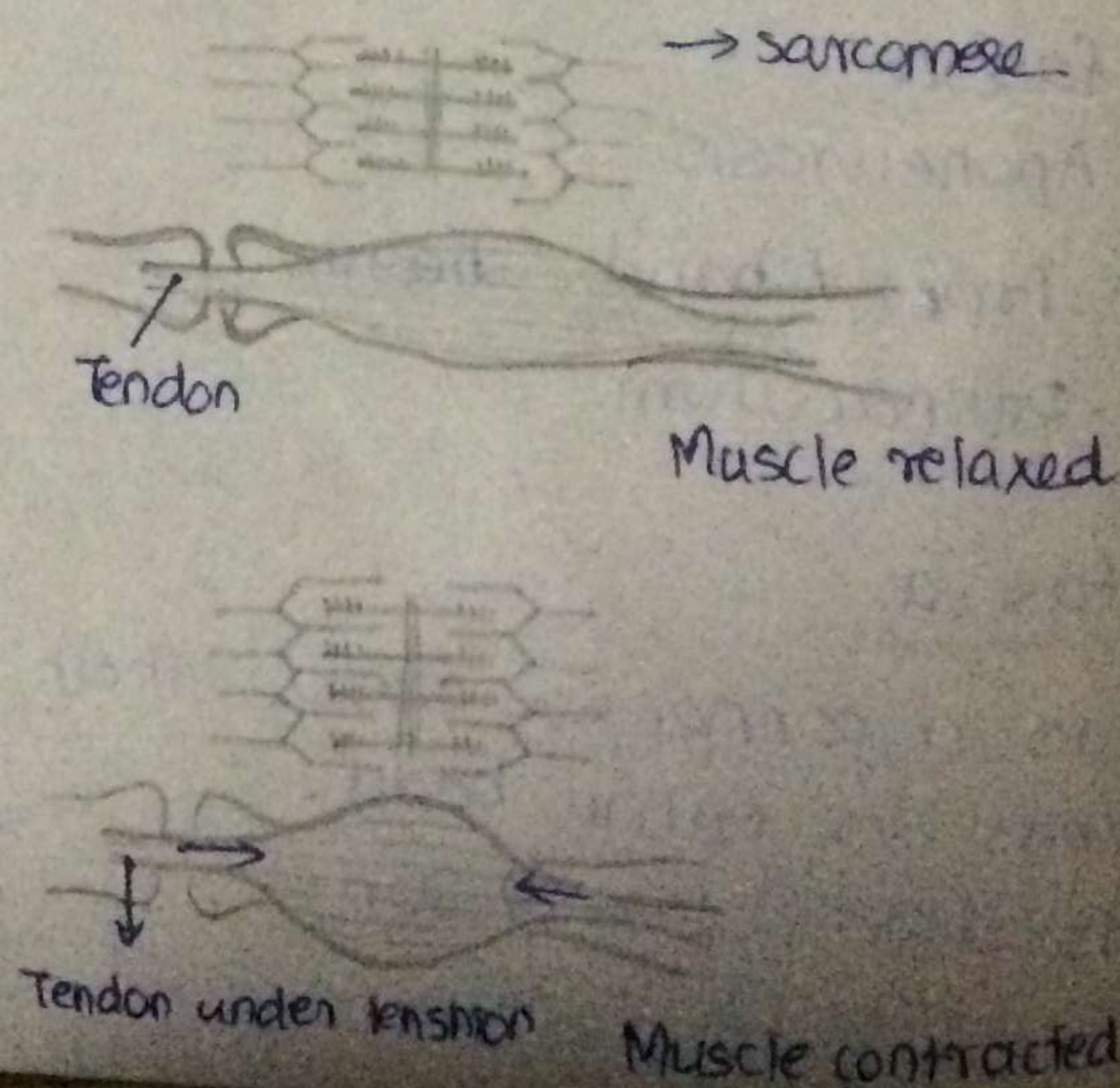
* When muscle contracts tendon gets stretched and tension increases.

* Whereas in eccentric contraction the tendon gets shortened (slack)

∴ passive tension is provided by tendon in concentric contraction and tension is zero or less during eccentric as well as relaxed position.

* But usually the passive tendon tension developed inside the muscle is provided by only the parallel elastic component. It is because the extensibility of the tendon is relatively small.

* Thus force generated is transmitted to the bony lever instead of stretching the tendon.



Muscle Function

I. Muscle tension

1) passive: Tension produced by parallel elastic component.

eg: collagen, elastin, epimysin, perimysin, endomysin.

2) Active: (M) contraction

ATP

cross bridge

Tension developed by the contractile element of the (M)

3) Total: (A)(P) divided.

1) Neural - motor unit frequency, magnitude of firing.

2) mechanical -

Isometric $\rightarrow$ length-tension relation.

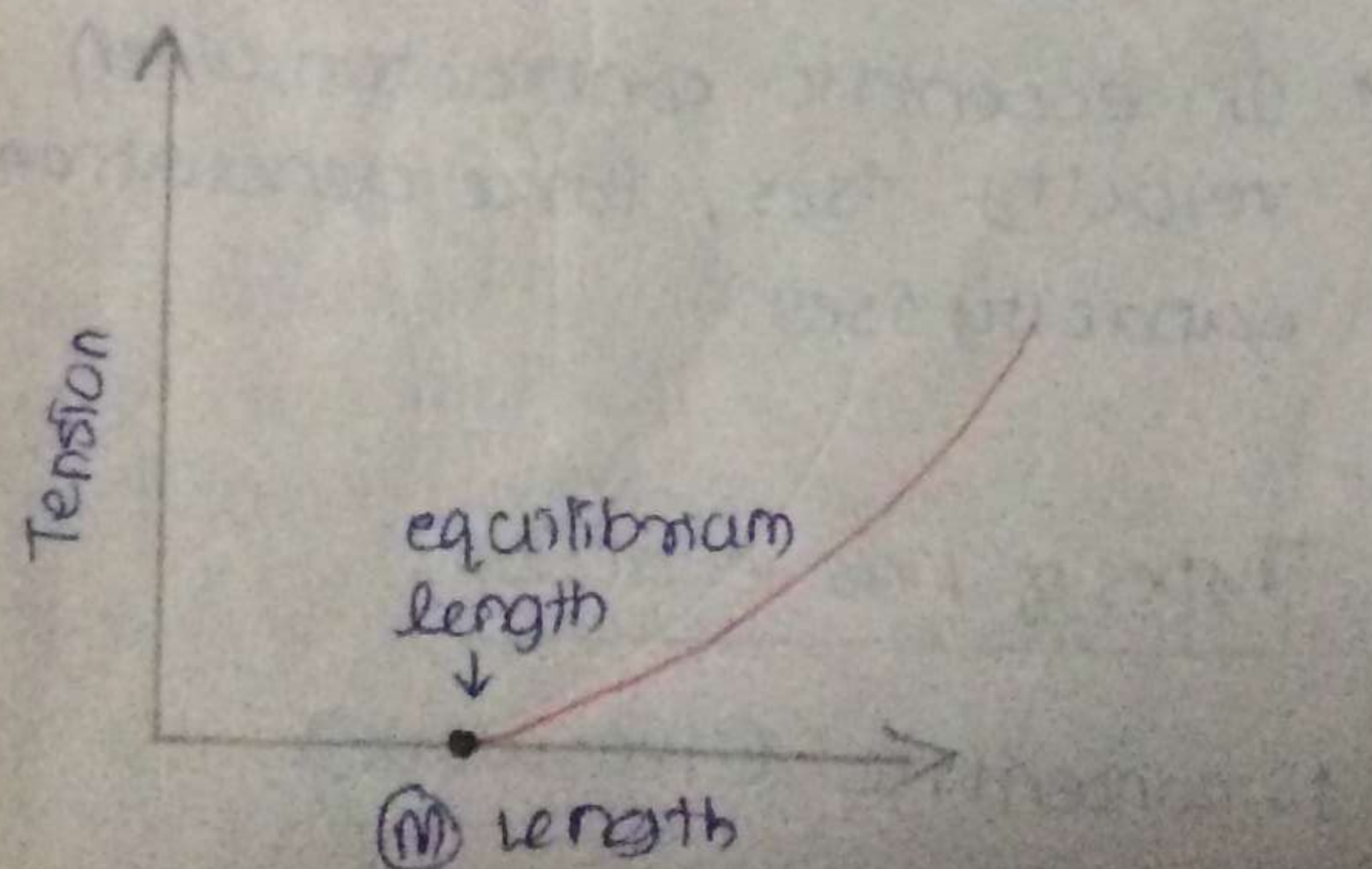
force - velocity relation.

* Tension developed in 11 el elastic component of (M).

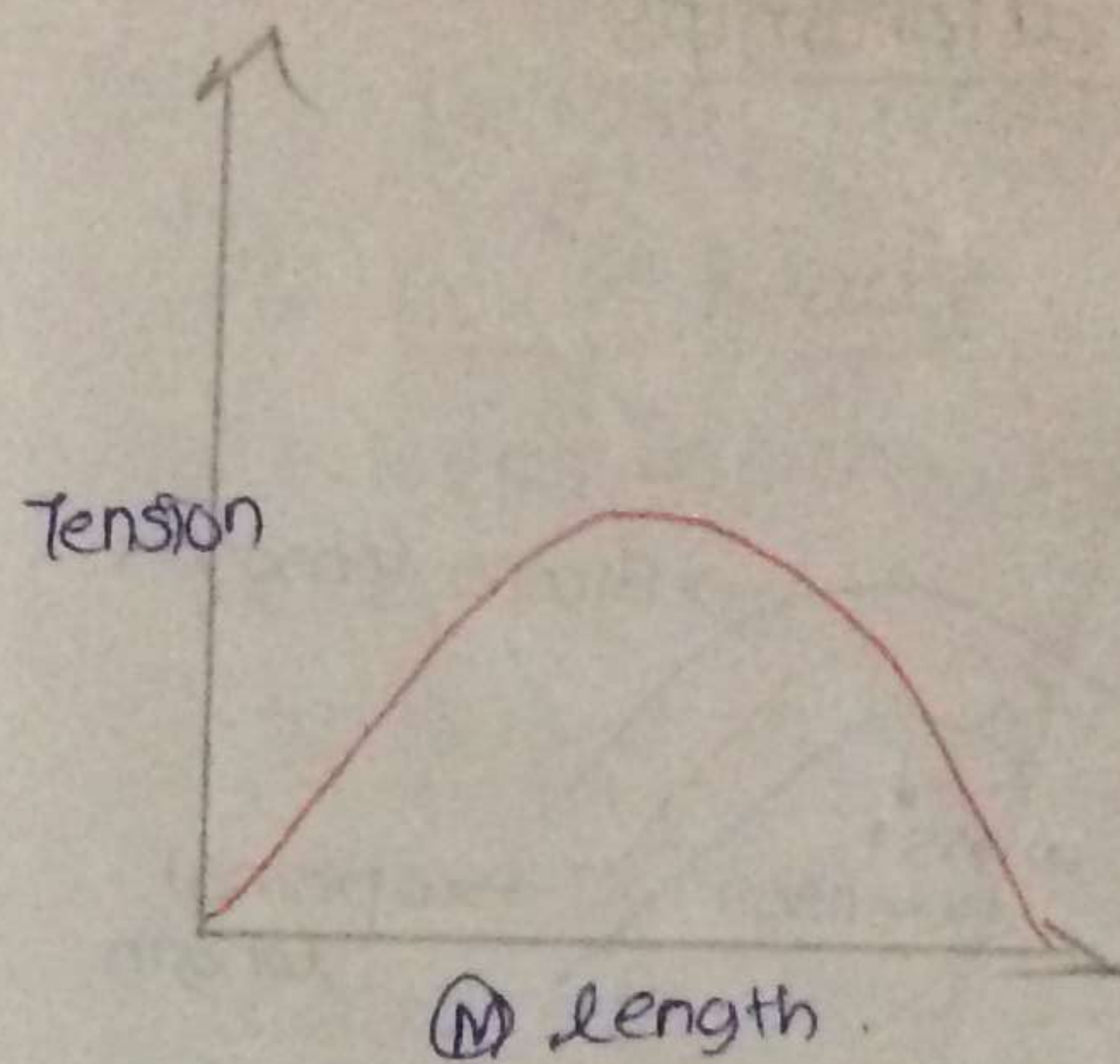
* It occurs due to lengthening of (M) beyond slack length of tissues or the stretching of (M).

* 11 el components added up tension developed in (M) when it lengthens.

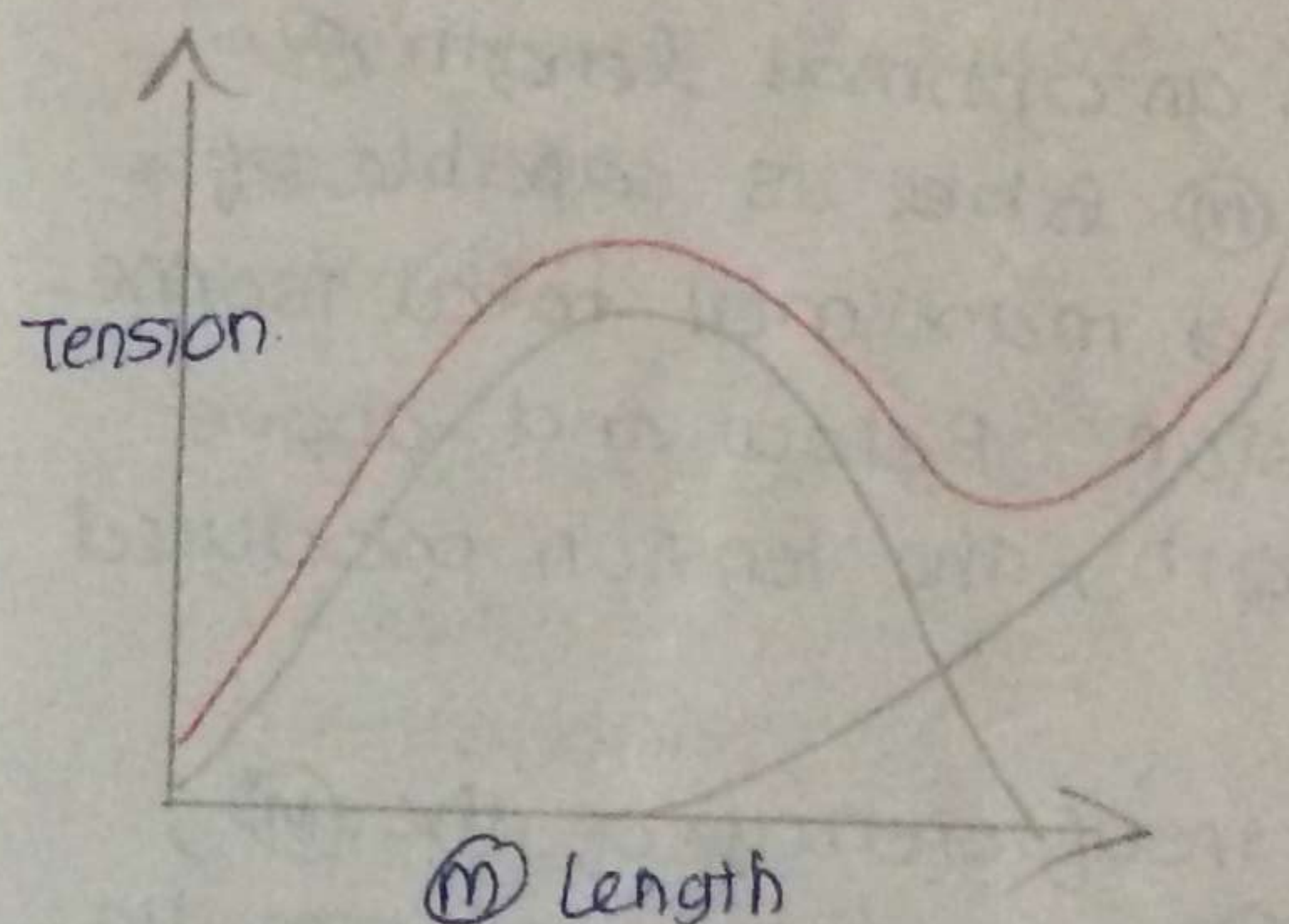
Passive tension



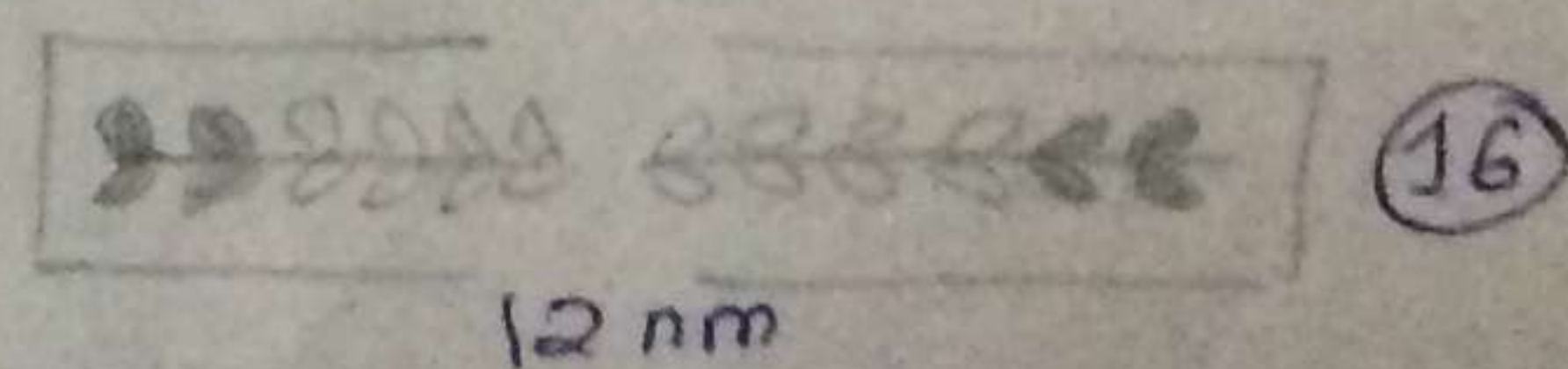
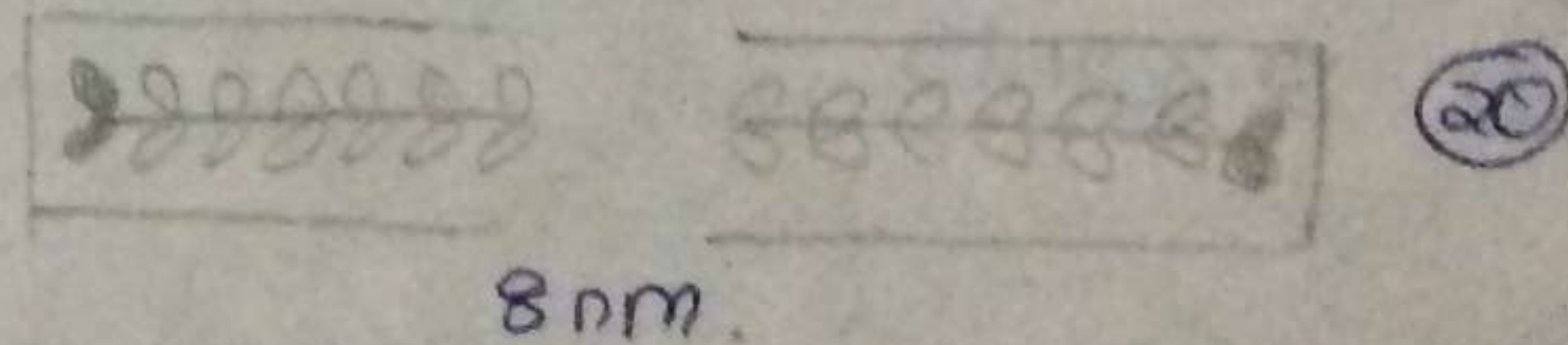
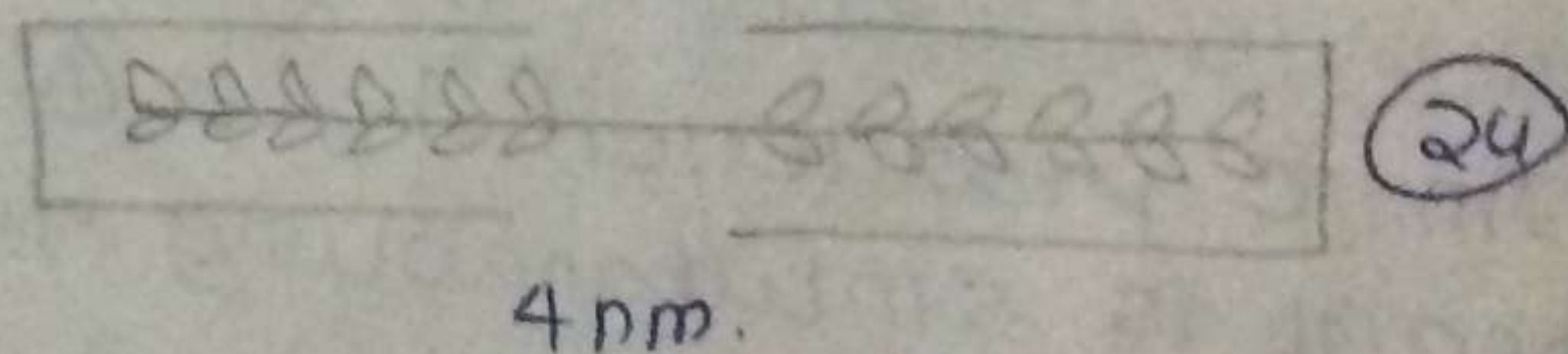
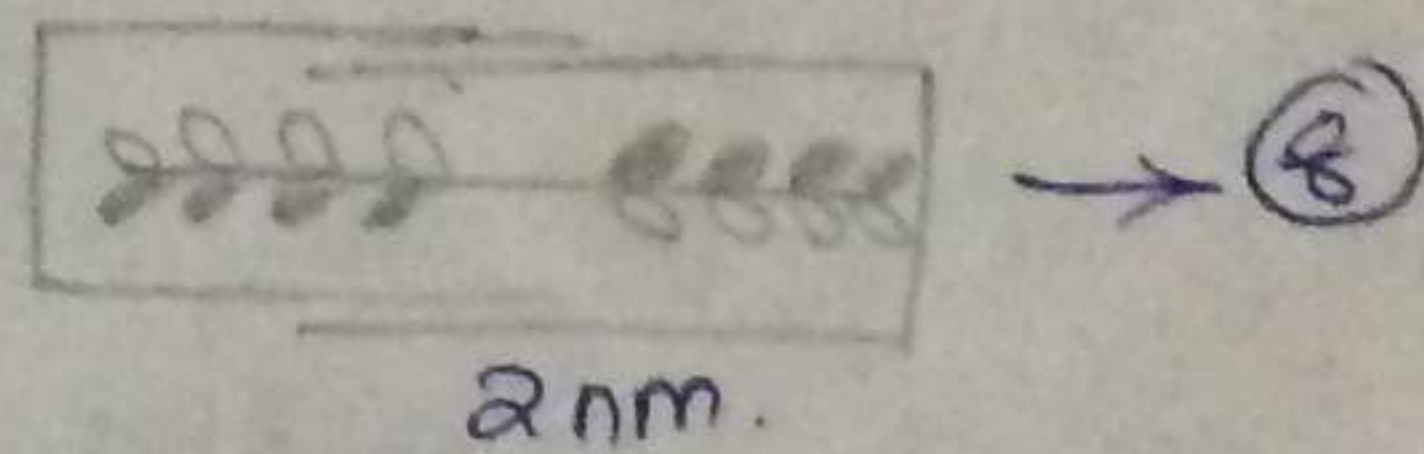
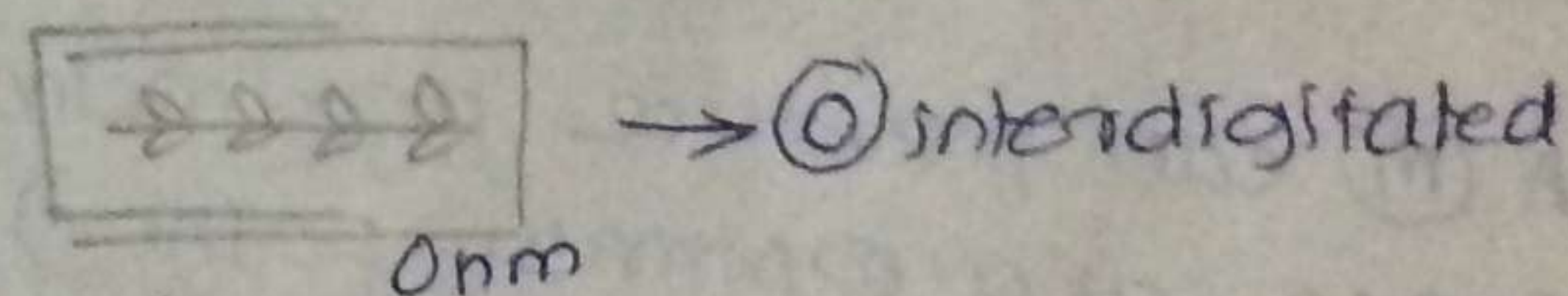
Active tension



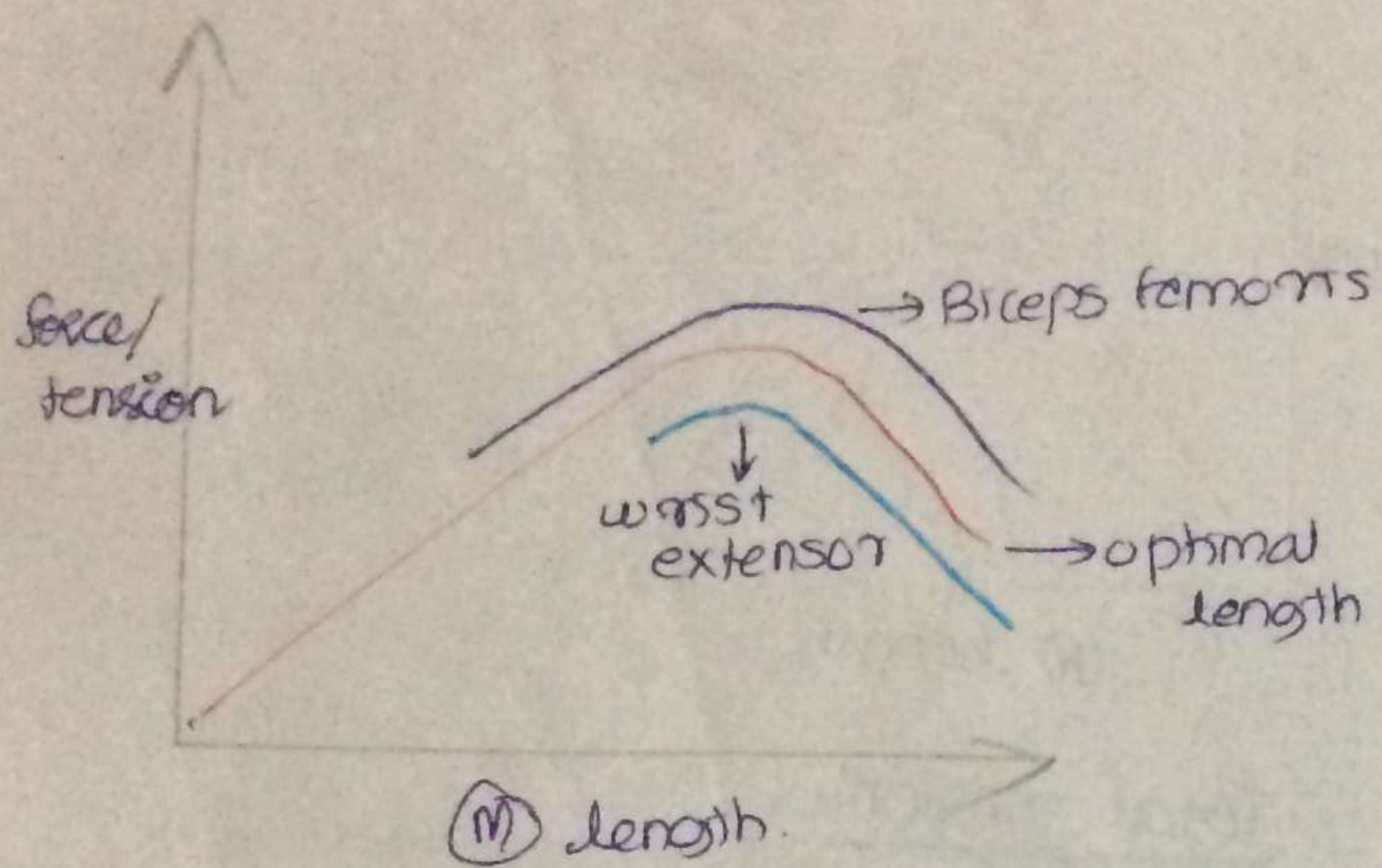
Total tension.



Isometric length-tension relationship of a muscle



Application of Isometric Length-Tension relationships



There is an optimal length (l_0) which a fibre is capable of developing maximal total isometric tension. Below and above the length, the tension produced is less.

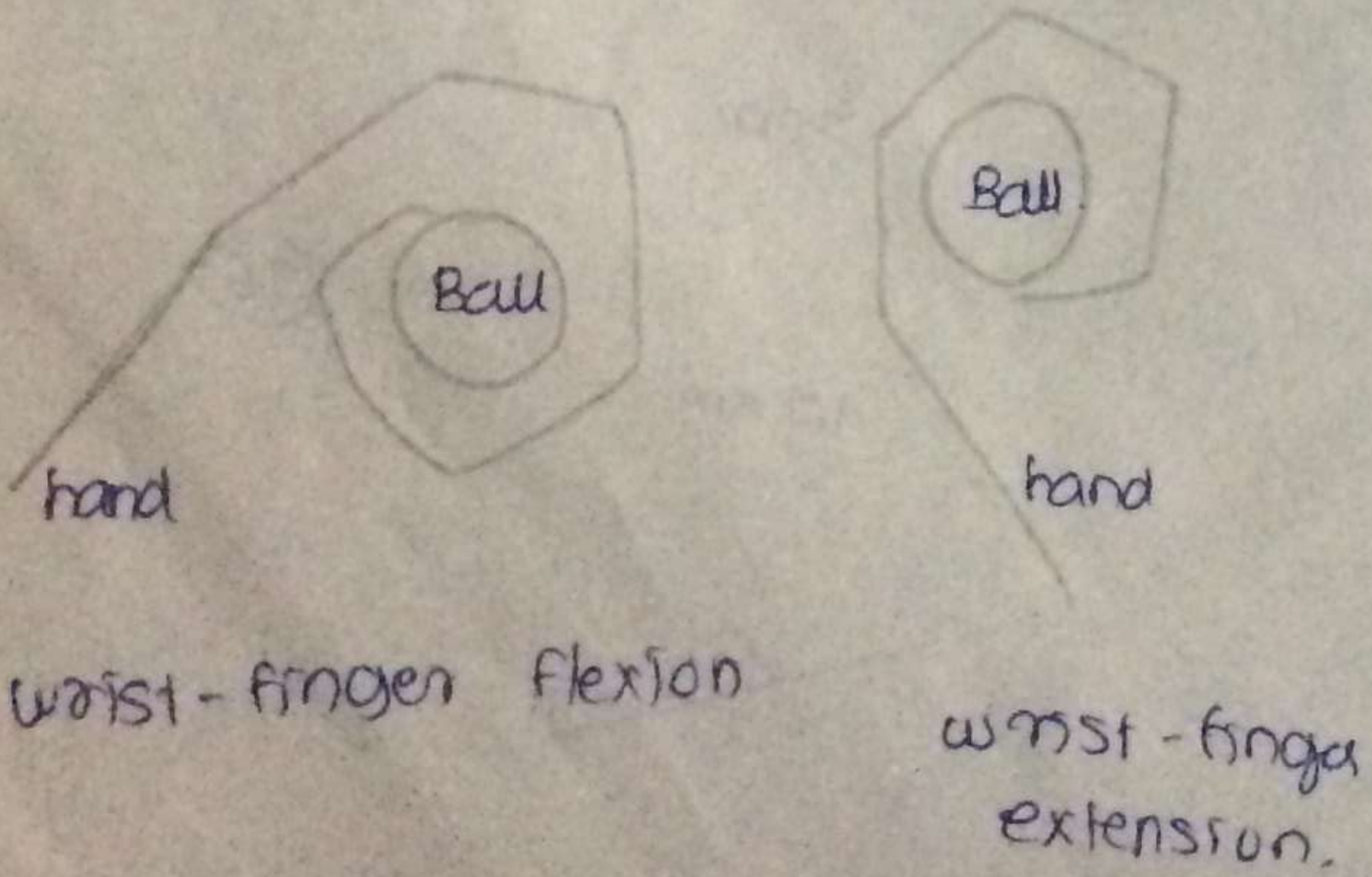
- * Active insufficiency of the (M) ,
2 joint (M) are recruited simulta-
neously, the force generation
capacity reduces.

eg: wrist, finger flexion-extension.

(A $\textcircled{M}$ can produce maximum force at a $\textcircled{P}$ primal point)

Application

Active Insufficiency - It is the $\downarrow$ se in τ production by a $\textcircled{M}$ when it is simultaneously recruited at 2 joints.

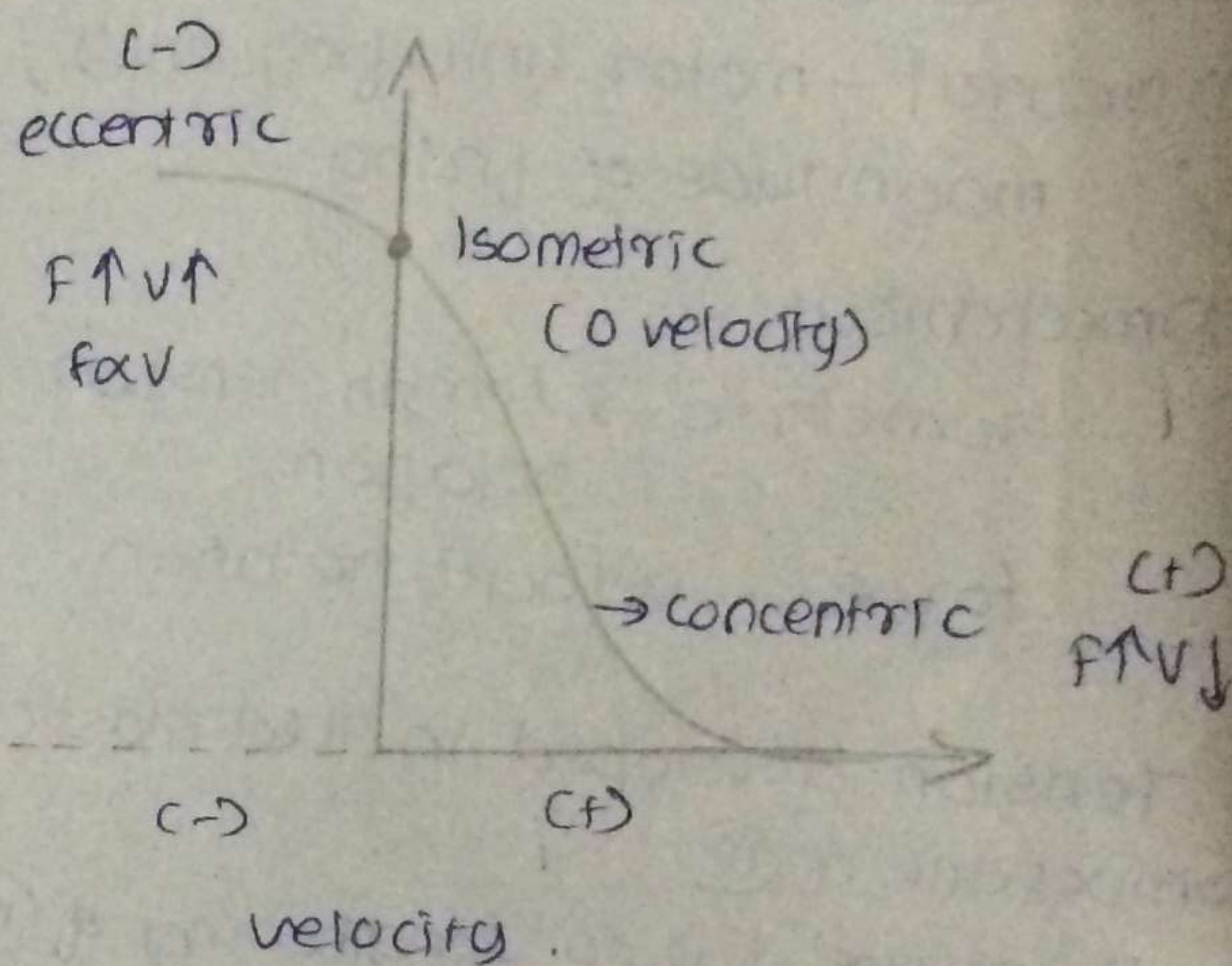


Force-velocity relationship

factors effecting active tension.

- * size of motor unit
- * no: and size of motor unit (M) fibre
- * No: of motor unit firing
- * sarcomere length.
- * fibre arrangement.
- * Type of (M) contraction
- * velocity.

Force-Velocity Relationship



- * The ability of a (M) to generate force, is affected by the speeds @ which the concentric or eccentric contraction occurs.
- * In concentric contraction when velocity ↑, force generation capacity ↓.
- * In eccentric contraction when velocity ↑, force generation capacity ↑.

Types of muscle action.

1. concentric (+ve work)

2) Isometric (zero work)

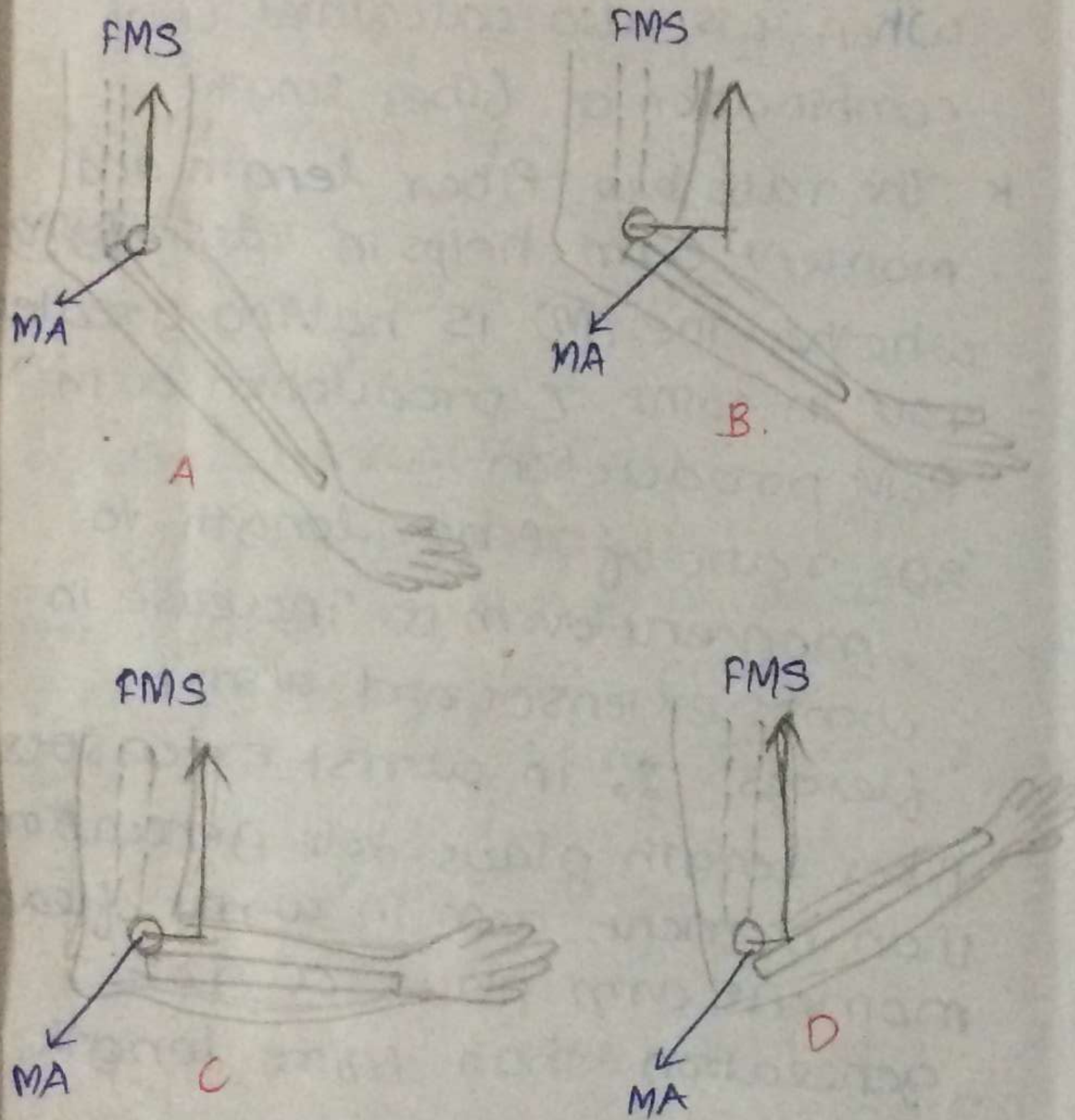
3) eccentric (-ve work)

4) Reverse work.

Classification based on production of τ

* when moment arm $\uparrow$ ses, the efficiency of (M) to produce or do work $\uparrow$ ses.

* either the τ generated by the (M) changes as moment arm of the (M) $\uparrow$ ses or $\downarrow$ ses.



Classification of muscle

1. Based on shape

2. Based on no: of heads

3. combination of location and function.

4. Based on action

5. Based on roll

6. Based on (M) architecture

7. Based on length of moment arm

8. Based on location.

1. Based on shape

Deltoid, rhomboids.

2. No: of heads

Triceps brachii, quadriceps, biceps.

3. Based on location and function.

extensor pollicis longus, extensor digitorum longus.

4. Based on action

* flexors - produce flexion

* extensors - produce extension

* abductors

* adductors

5. Based on roll

i) Agonist

ii) Antagonist

iii) synergids.

i) Agonists :-

* They are known as 1^o movers.

* main role is to produce a desired motion at a joint.

* If flexion is desired motion flexors are 1^o movers

* (M) that produce greatest amount of force is usually called agonist.

* eg: In elbow flexion, flexion is produced by biceps coracobrachialis. Here biceps produce greatest amount of force is usually called agonist.

* Hence, it is the prime mover.

ii) Antagonist :-

* (M) that acts against the agonist.

* eg: In elbow flexion, triceps brachii is the

antagonist.

- * Antagonist (M) are those (M) that produce ~~greatest amount of force~~, opposing joint $\angle$ to agonist.

iii) synergists

(M) that helps the action of agonist (M).

eg: In shoulder flexion biceps brachii act as synergists.

Here, deltoid is the agonist.

NOTE

No contraction :-

Here agonist and antagonist (M) work simultaneously. It is also called purposeful simultaneous contraction of agonist and antagonist.

- * It helps in increased stability.
- * Forms of synergists b/w (M)
- * It may help agonist directly to perform desired action or may assist indirectly by stabilizing a segment or preventing undesired action.

Based on (M) architecture

- * Some (M) may have $\uparrow$ ed PCSA and $\downarrow$ ed fiber length or some have $\downarrow$ ed PCSA and $\uparrow$ ed fiber length.
- * If PCSA is higher, role of (M) is force production.
- * If PCSA is $\downarrow$ ed, role of (M) is ROM production.
- * eg: In lower limb, extensor (M) (quadriceps) have $\uparrow$ ed PCSA and $\downarrow$ ed fiber length and hamstring has $\downarrow$ ed PCSA and increased fiber length.

$\therefore$ function of hamstring is ROM production, whereas quadriceps force generation.

- * soleus (plantar flexor) has increased PCSA.
- * Tibialis anterior have longer fiber length.

Based on Length of moment arm

- * Relationship of a (M) to the joint can also be used to classify (M).
- * Length of moment arm is important in determining joint $\angle$.
- * It is also used to determine ROM, when it is also calculated with combination of fiber length.
- * The ratio b/w fiber length and moment arm helps in identifying whether the (M) is having greater role in joint $\angle$ production or in ROM production.
eg: ratio of fiber length to moment arm is increase in wrist extensor and wrist flexors. In wrist extensors fiber length plays role generation than moment arm in wrist flexor moment arm plays a role generation than fiber length.

Factors Affecting (M) function

1. Type of joint and location of (M) attachment.

- * The ROM available to a joint varies with the type of joint.
- * Some joint allows all degrees of freedom whereas other joints allow only few degrees of freedom.
eg * Ball and socket joint - 3 degrees of freedom

- * pivot joint - 1 degree of freedom.
- * The joint type also determine whether the movement is flexion, extension, abduction, adduction.
- * The line of attachment also determine the (M) function.
- * anterior - flexors
- * posterior - extensors.
- * Lateral - abductors
- * medial - adductors.
- * The location of attachment of (M) whether it is near to joint also determine the (M) funcⁿ.
- * If it is near to the joint they provide mobility.
eg: Biceps.
- * If it is far to the joints, they provide stability.
eg: Brachioradialis

A (M) provide maximum joint stabilization at the point at which, its compressive (f_x) component is greater and rotatory component (f_y) is less.

2. NO. of joint

whether it is single joint or multijoint (M) that are recruited or activated is important in determining (M) function based on energy conservation.

Single joint

- * They are recruited for force and work production
- * They often acts in concentric and isometric contraction.

Multi joint

- * They are recruited to control

production in dynamic motion of body.

- * Their main funcⁿ is eccentric contraction.
- * Multijoint (M) are recruited during complex motion requiring movement across multiple joints.

eg: Biceps brachii

It is recruited during elbow flexion + supination during the motion of elbow flexion and supination (Complex mvmt) Biceps brachii which is a 2 joint (M) is recruited.

Brachialis

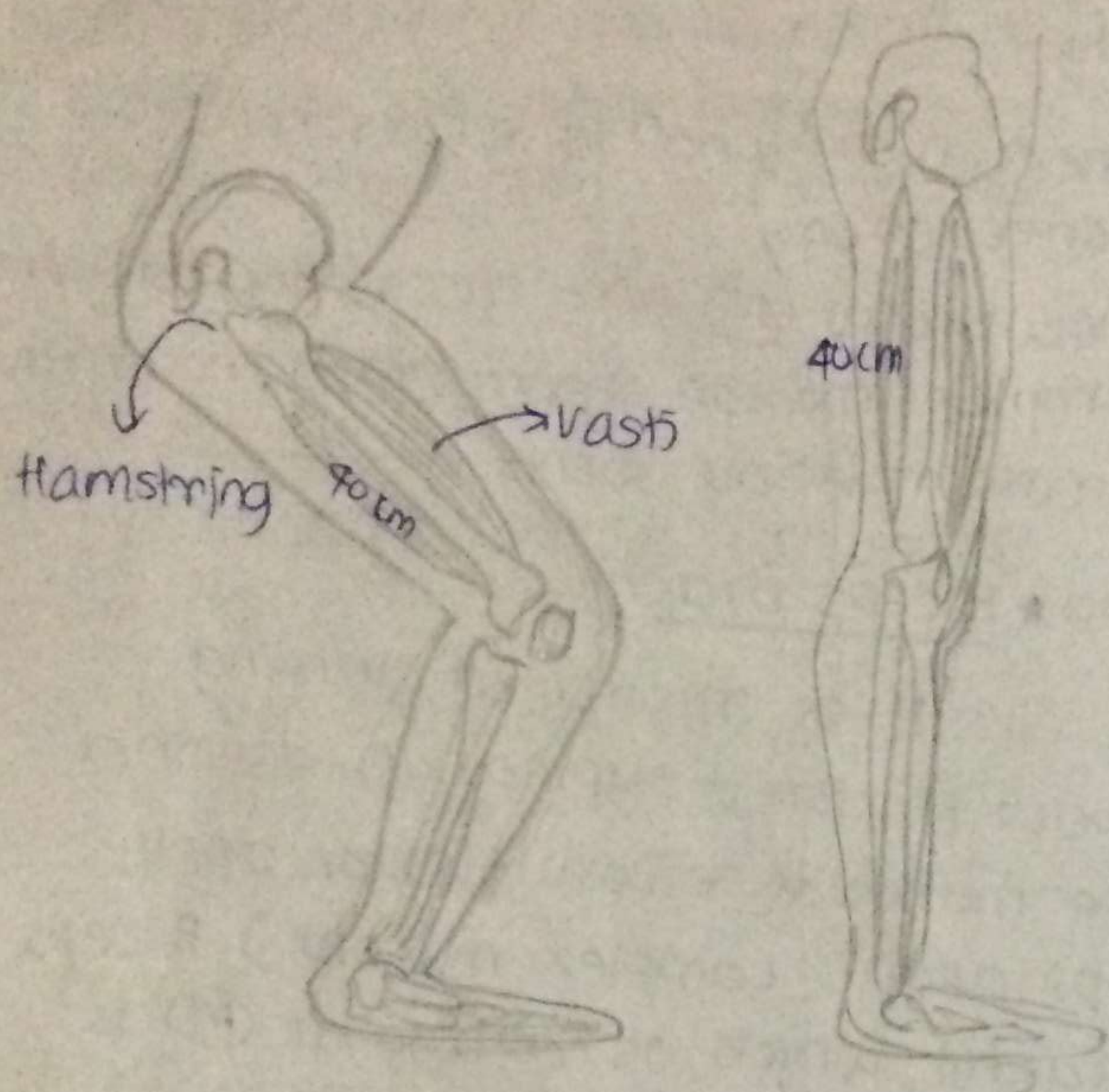
During elbow flexion in pronation (simple movement) brachialis (M) which is a single (M) is recruited.

If biceps is recruited during that movement, unnecessary supination occurs, that will require forearm pronator (M) to function to compensate bicep action.

Single joint (M) and multijoint (M) ~~are~~ may work together such a way that single joint (M) can assist in movement.

eg: standing up from a chair requires simultaneous knee and hip extension.

- * single joint knee extensor (M) assist multijoint hamstring with hip extension. Here, the knee extensors shorten and extend the femur that is a way help the hamstring (M).



3 Pa

3 Passive insufficiency.

It refers to inability of a multi-joint (M) to lengthen the degree that allows full range of motion of all joints it crosses simultaneously. The insufficient length in the passive (M) is called passive insufficiency.

eg 1: If a person's elbow is placed on a table with forearm in vertical position and if the relaxed hand drops forward into wrist flexion the fingers tends to extend.

* Extension of the fingers occurs because finger extensors are stretched over the flexed wrist and the insufficiently long to permit finger flexion.

eg 2: Hamstring muscles may limit full knee extension in hamstring are stretch over fully flexed hip.

eg 3: If a person's elbow is placed on a table with the forearm in vertical position and relaxed hand drops backward into wrist extension fingers extends to flex.

NOTE:

- * Usually 2 joint and multi-joint (M) exhibit passive insufficiency.
- * under normal conditions, 1 joint (M) do not exhibit passive insufficiency.
- * passive insufficiency can occur simultaneously with active insufficiency.

eg: when a person is attempting to grasp and reject with wrist in full flexion, person find typical to grasp the object by forming full flexion of fingers (active insufficiency).

Here, even though, the flexor (M) are unable to perform the full motion, the extensor group is stretched and lengthened. Thus the inactive finger extensors are passive lengthened and experiences passive insufficiency.

4. Sensory Receptors

Motor control is achieved through coordination of descending motor pathways from cortex, (M) action and a constant flow of sensory information.

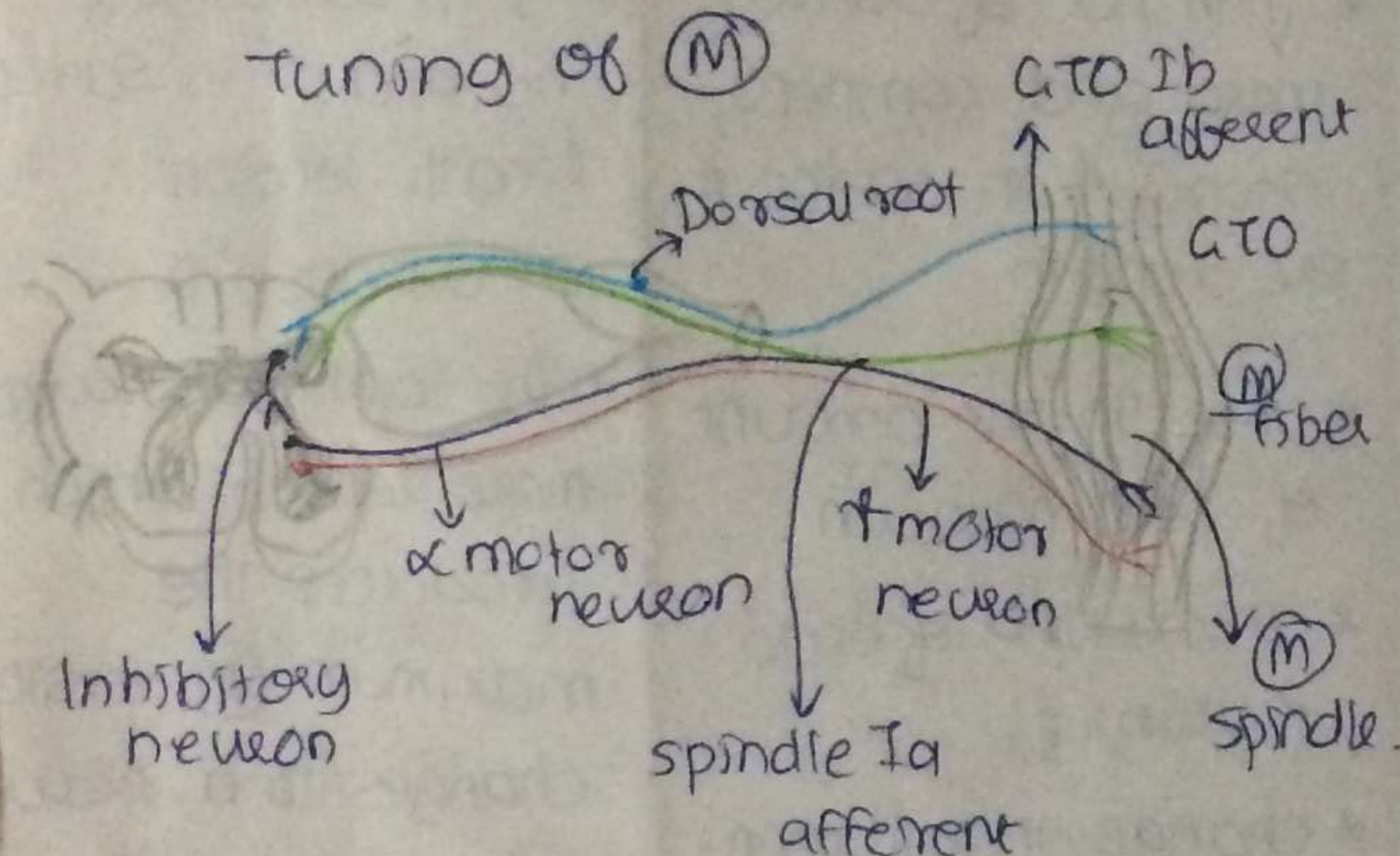
→ In (M) there are 2 important sensory receptors.

1. Golgi Tendon Organ (GTO)
2. (M) spindle.

GTO

- * location: In the tendon at MTJ.
- * function: sensitive to tension.
- * Activated by ~~active~~ either an active (M) contraction or by excessive passive stretch of (M).
- * Action:

When GTO are excited
↓
Type Ib afferent fibre
↓
CNS
↓
efferent fibres
↓
tuning of (M)



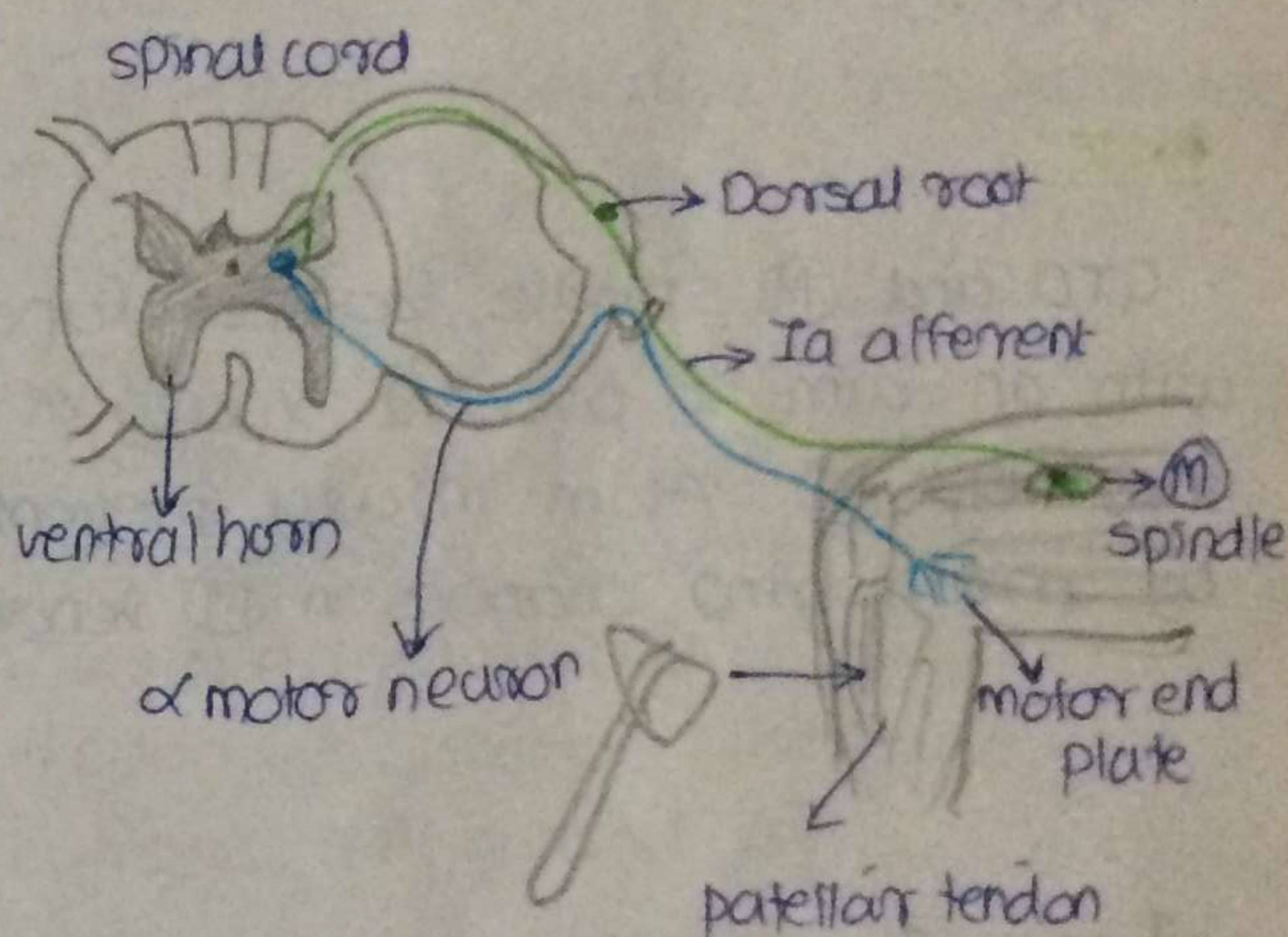
Muscle Spindle

- * It consists of 2-10 specialised (M) fibers and called intrafusal fibers
- * They are enclosed in a connective tissue sheath and are seen in the (M) belly.
- * function: Sensitive to length and velocity of lengthening of the (M) fiber either stretch of a (M).

* Action:

Muscle spindle
↓
Type Ia afferent fibers
↓
CNS
↓
efferent fibers
↓
α motor neuron
↓
tuning of (M)

- * They usually response to passive stretch of a (M).



special case:

- * when the (M) spindle reflex uses the CNS sense a message through γ-motor neuron to intrafusal (M) fibers in the (M) spindle which stretches the centre of the spindle slightly to ↑ the sensitivity.

Hence, spindle is one again able to respond to length change in a shortened (M).

When a (M) fiber shortens, spindle stops sending message because they are no longer stretched.

DTR

- * (M) spindle is responsible for sending the message to the (M) in which it lies to contract, when the tendon of the (M) is palpated with a hammer.
- * The quick stretch of the (M) caused by tapping the activates (M) spindle. Hence the (M) response to unexpected spindle message with a brief contraction. This response is called deep tendon reflex (DTR)
- * DTR is also known as (M) spindle reflex (MSR) or simply stretch reflex.

Note:

GTO and (M) spindle together function with an aim to prevent injury or they help (M) from injury prevention by monitoring changes in (M) length.

Feed-forward control

- * Control of the voluntary movement is not completely accomplished by GTO and (M) spindle, it is controlled by various integrated strategies.
- * One such strategy is feed-forward control.
- * It acts through sensory input from visual and auditory system to anticipate and control body movements. Either, feed forward control allows anticipatory control of the (M) action.

Effects of immobilization, Injury and aging.

Immobilization:

- * effects both the (M) structure and function.
- * Effects depends upon the immobilization position (shortend or lengthend),
- * % of fiber within the (M) and length of immobilization period.

Shortend position

Structural changes	Functional changes
* ↓ in no. of sarcomere and compensatory ↑ in sarcomere length. * ↑ in the amount of perimysium. * Thickening of endomysium. * change in collagen orientation, becomes more in circumflexion. * ↑ in connective tissue to (M) fibre ratio. * ↓ in (M) mass and (M) atrophy	* ↓ in sarcomere no and ↑ in sarcomere length tension relationship. * The optimal sarcomere length which produces the maximum (M) tension changes to a new length. * There occurs adaptive shortening of a (M) which hinders (M) to a return to normal typical function immediately after immobilization period. * The adaptive shortening resists lengthening and hence ↓ in joint ROM. * ↑ in stiffness to passive stretch.

Length

structure

- * ↑ in sarcomere length in sarcomere
- * ↑ in sarcomere length and perimysium
- * There is hypertrophy by (M) atrophy

Solution

How to

- * 18 m of the
- * The can be
- * Ho
- * greater movement
- * frequency to one
- * If do
- * intensity and passive sarcomere due to
- * But it
- * It leads
- * (M) contraction

Lengthening position:

Structural changes

- * ↑ in no. of sarcomere and ↓ in sarcomere length.
- * ↑ in endomysial and perimysial C.T
- * There is a (M) hypertrophy followed by (M) atrophy.

Functional changes

- * There is ↑ in maximum tension generation, length-tension curve is displaced, passive tension remains almost same.

Solutions:

→ How to prevent problems:

- * 18 min of daily ROM activity out of the cast.
- * The -ve effects of immobilization can be minimised in shortend position.
- * However if immobilization is greater than 3 weeks, the benefits of movement will not be apparent, if frequency of movement is reduced to one per minute.
- * If daily movement is not possible, intensive mobilization with active and passive motion can reverse the sarcomere no and C.T alterations due to immobilization.
- * But if mobilization is aggressive, it leads to injury especially with (M) connective tissues.

Injury

→ 3 types of injuries can occur

- i) Overuse injury
- ii) (M) strain injury
- iii) Eccentric - exercise induced (M) injury.
- * DOMS - Delayed Onset (M) Soreness.

1. Overuse injury

- * The most common reason for the overuse injury is repetitive trauma.
 - * It occurs because there is not enough time for the complete repair of tissue.
 - * Subsequent repetitive force leads to microtrauma, which in turn triggers the inflammatory process and results in swelling.
 - * The tissue most affected by the overuse injury is MTJ.
 - * Overuse injury may occur in tendon, ligament, cartilage, bursae, nerve and (M).
- Inflammation of Bursae - Bursitis
- Nerve - compression injuries.

2. (M) strain injury

It occurs from a single high force contraction of a (M) when it is lengthened by an external force.

eg: Body weight.

- * Here the (M) fails at MTJ.
- * There is localised bleeding, inflammation (swelling, redness, pain).

3. Eccentric - Exercise (M) induced injury

- * Eccentric exercise induces injury to the (M), such injuries are commonly called **DOMS**.
- * DOMS is the pain and stiffness feel in a (M) several hours to days after high velocity exercise.
- * It usually occurs 24 to 72 hours after exercise and reaches a peak 2-4 days after the exercise.
- * DOMS usually occurs in (M) performing eccentric exercise and not in the concentric.
- * It may be a result of mechanical strain in the (M) fibres.
- * There is evidence for deformation of the Z line, collagen breakdown and for other connective tissue changes.

Aging

- Aging effects the (M) in 2 ways.
- i) fibre no: and fibre type change
 - ii) connective tissue change.

1. Fiber no: and fiber type change

- * As a person ages, skeletal (M) strength ↓ as a result of sarcopenia (loss of (M) mass)
- * Sarcopenia occurs as a result of loss of (M) fibers and ↓ in the size of existing fibers.

* It is commonly believed that no: and size of type II fibre decrease and type I fibre ↑ as age proceeds in type I fiber.

* In elderly people who are active fiber type and no: is maintained to a considerable level.

* This is also ↓ in the no: of motor units.

2. connective tissue changes

- * Aging ↑ the amount of C-T (endomysium, epimysium, perimysium)
- * The ↑ed amount of CT results in ↓ ROM and ↑ (M) stiffness.
- * "All this changes in the (M) result in ↓ (M) strength and more importantly a loss in (M) power. The loss of (M) power is the main contributing factor for falls in elderly."